

HALIÇ UNIVERSITY
Department of Computer Engineering
Computer Networks — Lab 3 Report

LAB 3

Topic: Wireless Network Simulation

INET Framework Wireless Tutorial — inet.omnetpp.org

Barkın Kocatepe

Student ID: 23091440020

Section: 1

Date: 05/05/2026

Talat Doğan Evcı

Student ID: 22091000302

Section: 1

Date: 05/05/2026

1. Selected Topic & Tutorial

Topic: Wireless

Tutorial Name: INET Framework Wireless Tutorial

Source: <https://inet.omnetpp.org/docs/tutorials/wireless/doc/index.html>

This tutorial introduces wireless network simulation using OMNeT++ and the INET Framework. Starting from two hosts with a basic radio model, each step progressively adds realism: multi-hop routing, CSMA MAC, energy models, node mobility, AODV ad-hoc routing, physical obstacles, and increasingly accurate radio propagation models.

2. Simulation Scenario Description

The tutorial builds a single wireless network scenario incrementally across 14 steps. The core setup throughout is:

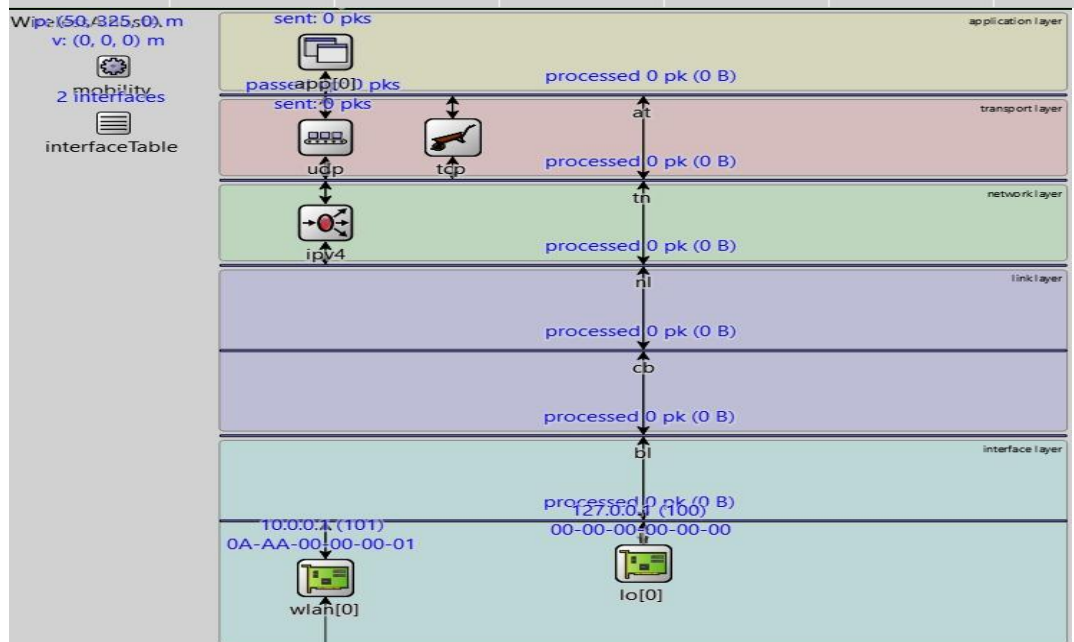
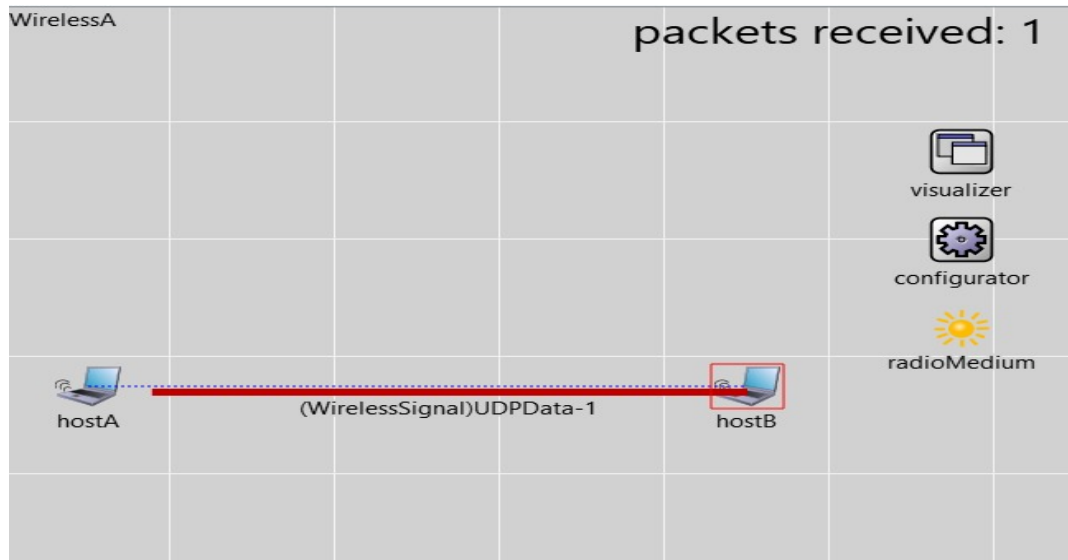
- Two endpoints: hostA (sender) and hostB (receiver)
- Two relay nodes: R1 and R2 positioned between the hosts
- A UDP application on hostA sending constant-bitrate traffic to hostB
- Communication range initially set to 500m (ideal), then reduced to 250m in Step 3

Each step modifies or extends the previous configuration by editing the omnetpp.ini file, introducing new INET modules (visualizers, MAC protocols, routing, mobility, energy, radio models) to progressively approach a realistic wireless network simulation.

3. Step-by-Step Simulation Walkthrough

The following 14 steps document the INET Wireless Tutorial in full. Each step describes what changes from the previous configuration, what to observe, and includes a placeholder for your screenshot.

Step 1	Two Hosts Communicating Wirelessly
The simplest possible wireless scenario: two hosts (hostA and hostB) placed 400m apart, each equipped with an 802.11 radio using the UnitDiskRadio model. HostA runs a UDP application sending 1000-byte packets to hostB at a rate of ~800 kbps. There is no MAC protocol yet — only a single IdealWirelessNic. The goal is to verify basic packet delivery and understand how INET structures a minimal wireless network in the NED file.	



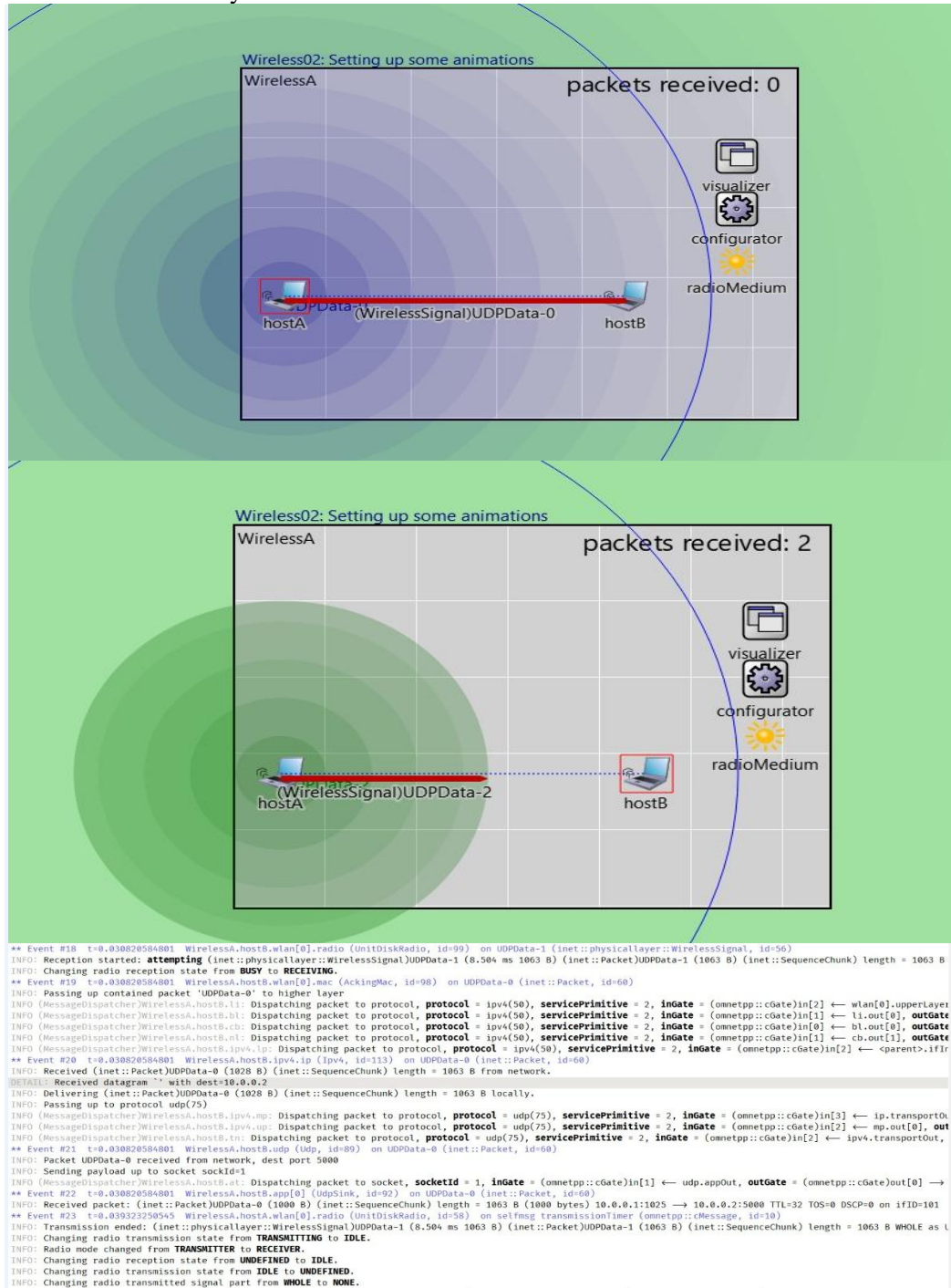
```

** Event #19 t=0.030820584801 WirelessA.hostB.wlan[0].radio (UnitDiskRadio, id=99) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=42065)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.584 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #19 t=0.030820584801 WirelessA.hostB.wlan[0].mac (AckingMac, id=98) on UDPData-0 (inet::Packet, id=42069)
INFO: Passing up contained packet 'UDPData-0' to higher layer
INFO (MessageDispatcher)WirelessA.hostB.l1: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessA.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← l1.out[0], outGate
INFO (MessageDispatcher)WirelessA.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessA.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessA.hostB.ipv4.ip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← parent>.ifir
** Event #20 t=0.030820584801 WirelessA.hostB.ipv4.ip (Ipv4, id=113) on UDPData-0 (inet::Packet, id=42069)
INFO: Received (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram "" with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessA.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessA.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessA.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #21 t=0.030820584801 WirelessA.hostB.udp (Udp, id=89) on UDPData-0 (inet::Packet, id=42069)
INFO: Packet UDPData-0 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessA.hostB.udp: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appout, outGate = (omnetpp::cGate)out[0] →
** Event #22 t=0.030820584801 WirelessA.hostB.app[0] (UdpSink, id=92) on UDPData-0 (inet::Packet, id=42069)
INFO: Received packet: (inet::Packet)UDPData-0 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
** Event #23 t=0.039323250945 WirelessA.hostA.wlan[0].radio (UnitDiskRadio, id=58) on selfmsg transmissionTimer (omnetpp::cMessage, id=42019)
INFO: Transmission ended: (inet::physicalLayer::WirelessSignal)UDPData-1 (8.584 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B WHOLE as l
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.

```

Step 2 Setting Up Some Animations

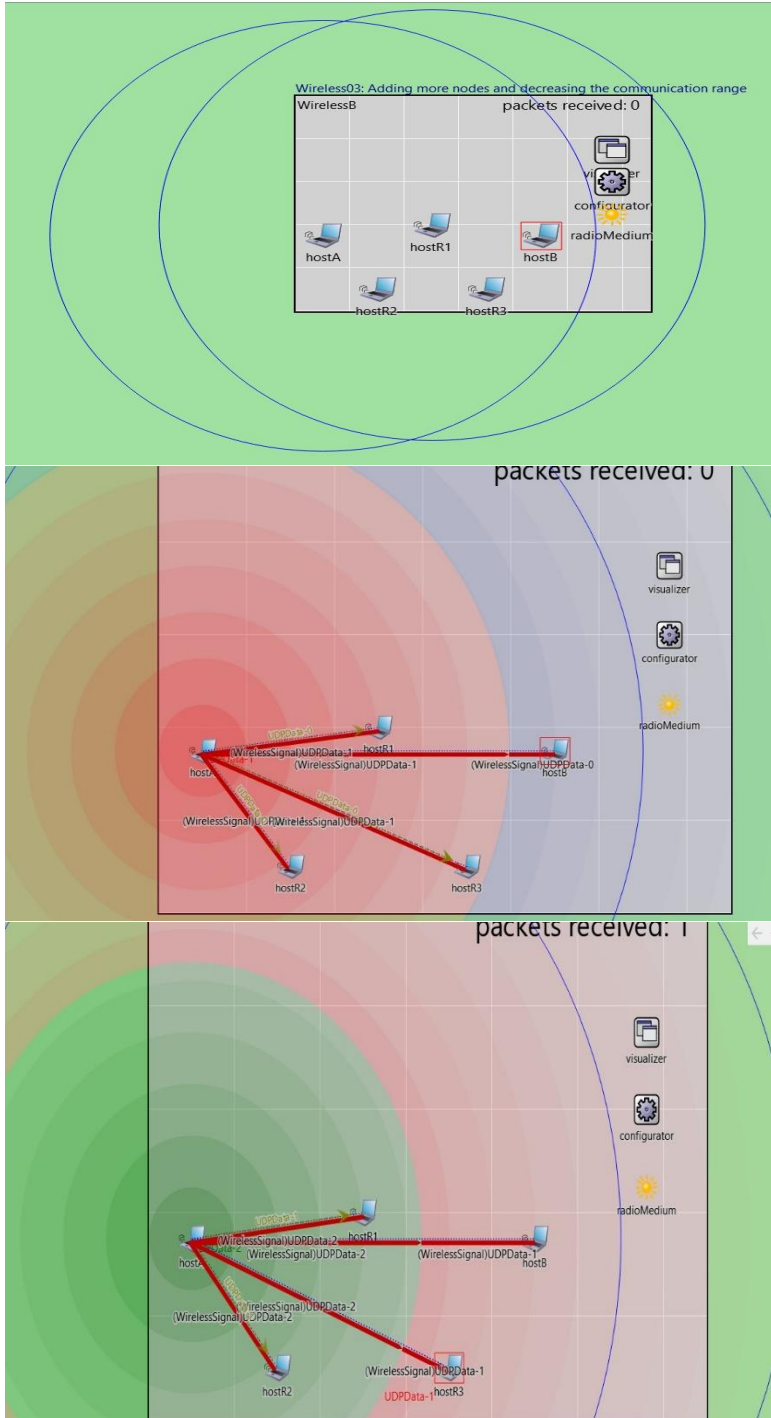
This step adds visualizers to make the simulation easier to understand visually. An `IntegratedCanvasVisualizer` is introduced, enabling data link activity arrows and physical link range circles around each node. The radio range is shown as a filled circle. This step does not change any functional network behaviour — it is purely about improving the visual feedback of the simulation for observation and analysis.



Step 3

Adding More Nodes and Decreasing the Communication Range

Two relay nodes (R1 and R2) are added between hostA and hostB, and the communication range is reduced to 250m so that hostA and hostB can no longer reach each other directly. Packets must now hop through R1 then R2. However, since there is still no routing or MAC protocol, this step demonstrates that packets are NOT yet successfully delivered — setting up the motivation for Steps 4 onwards.



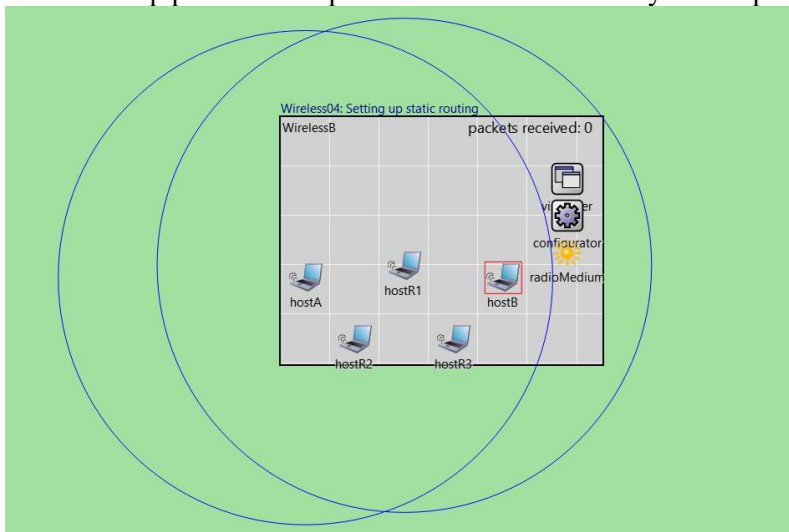

```

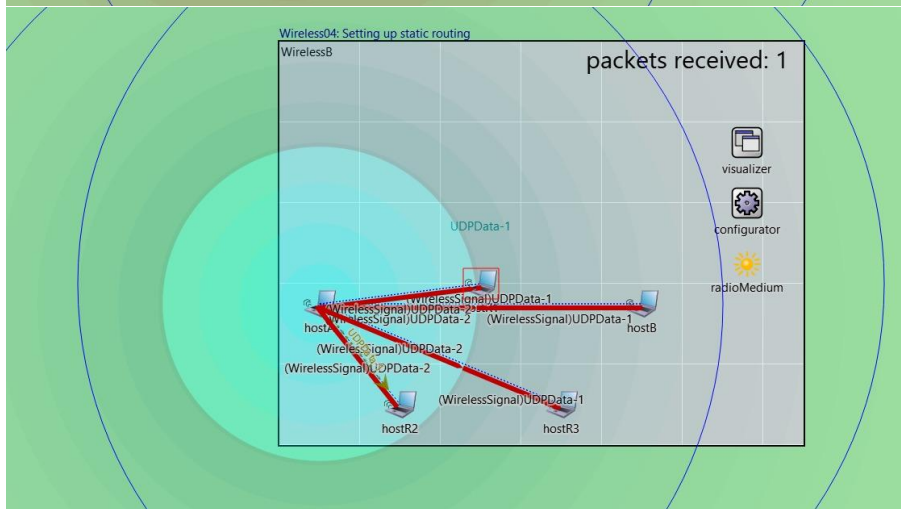
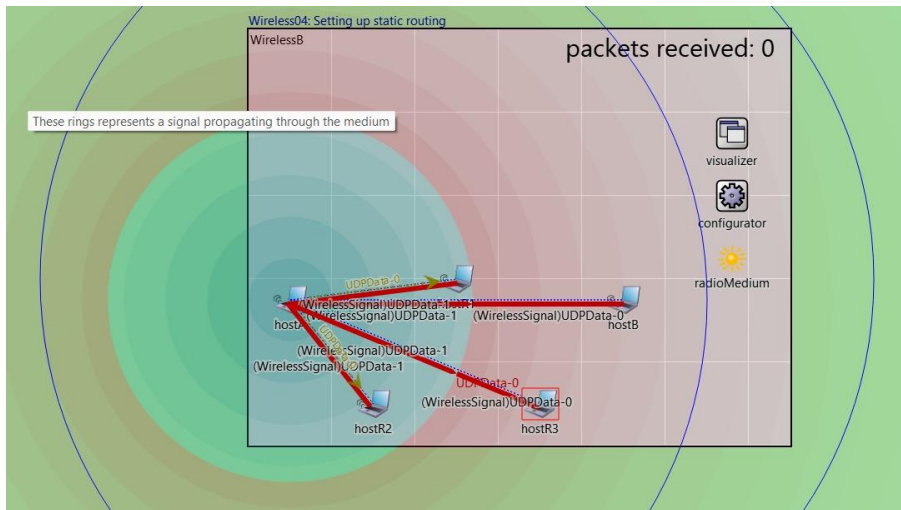
** Event #20 t=0.030819784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on selfmsg receptionTimer (omnetpp::cMessage, id=115)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #21 t=0.030819784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=126)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #22 t=0.030819784508 WirelessB.hostR2.wlan[0].mac (AckingMac, id=181) on UDPData-0 (inet::Packet, id=135)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #23 t=0.030819922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on selfmsg receptionTimer (omnetpp::cMessage, id=116)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #24 t=0.030819922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=124)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #25 t=0.030819922865 WirelessB.hostR1.wlan[0].mac (AckingMac, id=141) on UDPData-0 (inet::Packet, id=138)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #26 t=0.030820334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on selfmsg receptionTimer (omnetpp::cMessage, id=117)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #27 t=0.030820334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=128)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #28 t=0.030820334628 WirelessB.hostR3.wlan[0].mac (AckingMac, id=221) on UDPData-0 (inet::Packet, id=141)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #29 t=0.030820584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on selfmsg receptionTimer (omnetpp::cMessage, id=118)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #30 t=0.030820584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=122)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #31 t=0.030820584801 WirelessB.hostB.wlan[0].mac (AckingMac, id=101) on UDPData-0 (inet::Packet, id=144)
INFO: Passing up contained packet 'UDPData-0' to higher layer
INFO (MessageDispatcher)WirelessB.hostB.li: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← li.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessB.hostB.ipv4.ip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.ifIr
** Event #32 t=0.030820584801 WirelessB.hostB.ipv4.ip (Ipv4, id=116) on UDPData-0 (inet::Packet, id=144)
INFO: Received (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram "" with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessB.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessB.hostB.ipv4.up: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessB.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #33 t=0.030820584801 WirelessB.hostB.udp (Udp, id=92) on UDPData-0 (inet::Packet, id=144)
INFO: Packet UDPData-0 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessB.hostB.at: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appOut, outGate = (omnetpp::cGate)out[0] →
** Event #34 t=0.030820584801 WirelessB.hostB.app[0] (UdpSink, id=95) on UDPData-0 (inet::Packet, id=144)
INFO: Received packet: (inet::Packet)UDPData-0 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
** Event #35 t=0.033928880414 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=21)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut,

```

Step 4 Setting Up Static Routing

An IPv4NetworkConfigurator is added to automatically assign IP addresses and install static routes on all nodes. Each node is now a proper IP host. The configurator computes shortest-path routes based on the physical topology, so hostA -> R1 -> R2 -> hostB is established. UDP packets now successfully traverse the multi-hop path. This step introduces the network layer on top of the existing physical layer.





```

** Event #20 t=0.030819784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on selfmsg receptionTimer (omnetpp::cMessage, id=155)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #21 t=0.030819784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=166)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #22 t=0.030819784508 WirelessB.hostR2.wlan[0].mac (AckingMac, id=181) on UDPData-0 (inet::Packet, id=175)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #23 t=0.030819922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on selfmsg receptionTimer (omnetpp::cMessage, id=156)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #24 t=0.030819922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=164)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #25 t=0.030819922865 WirelessB.hostR1.wlan[0].mac (AckingMac, id=141) on UDPData-0 (inet::Packet, id=178)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #26 t=0.030820334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on selfmsg receptionTimer (omnetpp::cMessage, id=157)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #27 t=0.030820334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=168)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
** Event #28 t=0.030820334628 WirelessB.hostR3.wlan[0].mac (AckingMac, id=221) on UDPData-0 (inet::Packet, id=181)
INFO: Frame 'UDPData-0' not destined to us, discarding
** Event #29 t=0.030820584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on selfmsg receptionTimer (omnetpp::cMessage, id=158)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-0 (8.504 ms 1063 B) (inet::Packet)UDPData-0 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-0 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #30 t=0.030820584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on UDPData-1 (inet::physicalLayer::WirelessSignal, id=162)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #31 t=0.030820584801 WirelessB.hostB.wlan[0].mac (AckingMac, id=101) on UDPData-0 (inet::Packet, id=184)
INFO: Passing up contained packet 'UDPData-0' to higher layer
INFO (MessageDispatcher)WirelessB.hostB.l1: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← li.out[0], outGate = (omnetpp::cGate)out[1]
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate = (omnetpp::cGate)out[0]
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate = (omnetpp::cGate)out[1]
INFO (MessageDispatcher)WirelessB.hostB.ipv4.lp: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.l1

```



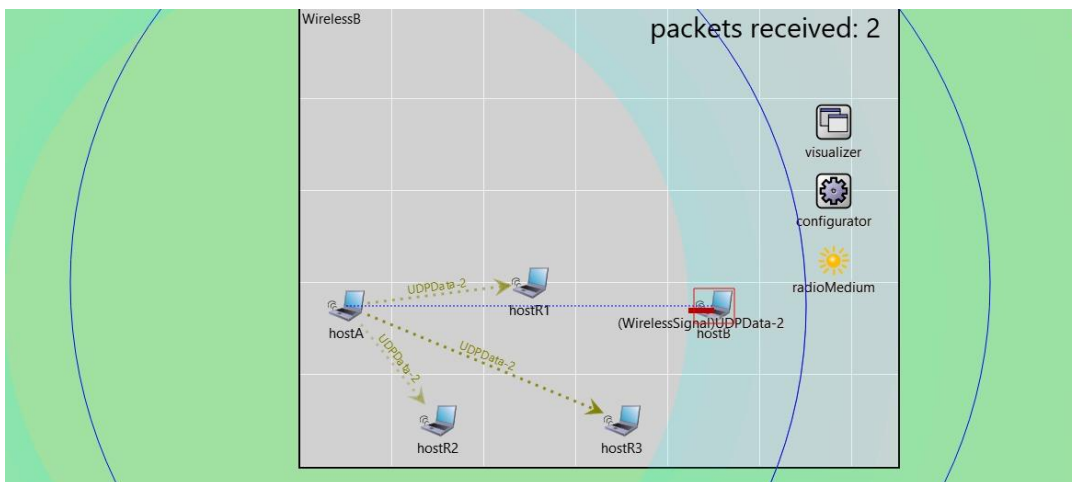
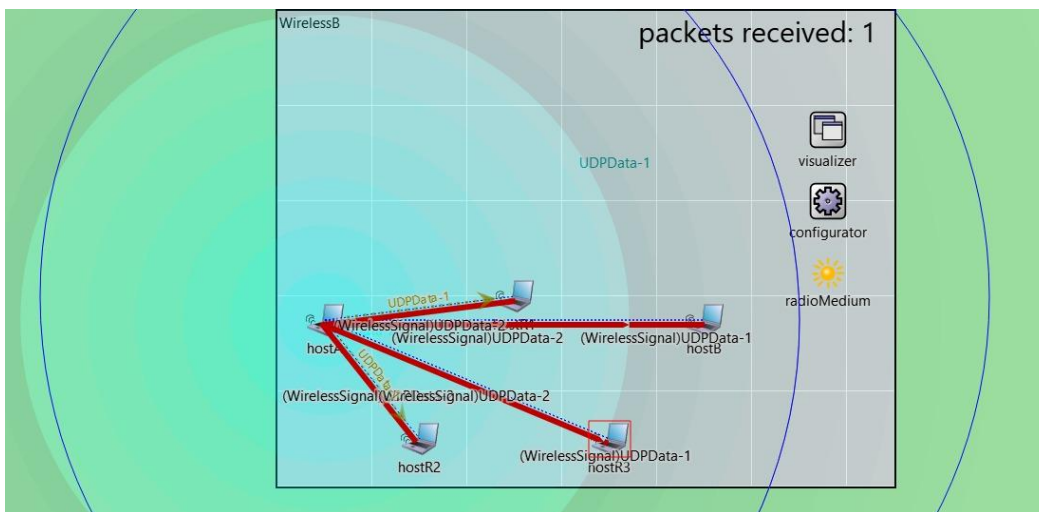
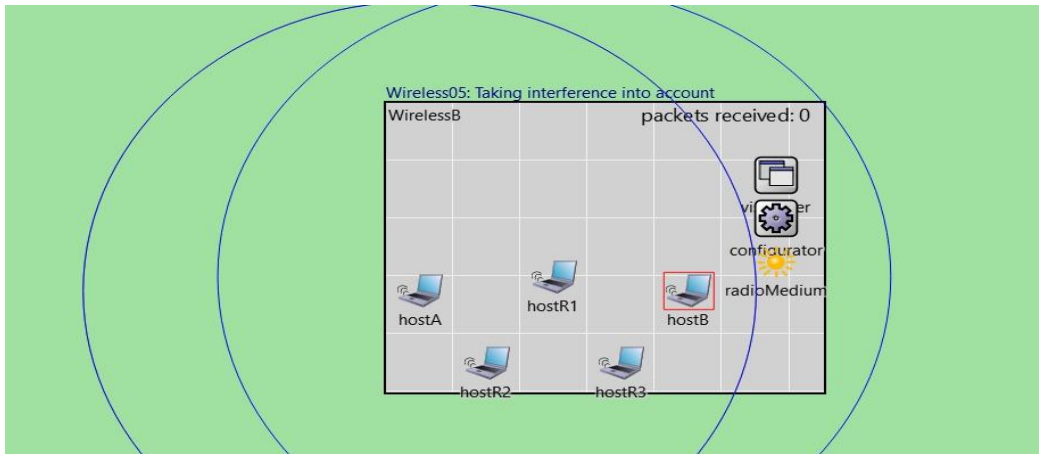
```

** Event #32 t=0.030820584801 WirelessB.hostB.ipv4.ip (IPv4, id=116) on UDPData-0 (inet::Packet, id=184)
INFO: Received (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram '' with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessB.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessB.hostB.ipv4.up: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessB.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #33 t=0.030820584801 WirelessB.hostB.udp (Udp, id=92) on UDPData-0 (inet::Packet, id=184)
INFO: Packet UDPData-0 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessB.hostB.at: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appOut, outGate = (omnetpp::cGate)out[0] →
** Event #34 t=0.030820584801 WirelessB.hostB.app[0] (UdpSink, id=95) on UDPData-0 (inet::Packet, id=184)
INFO: Received packet: (inet::Packet)UDPData-0 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
** Event #35 t=0.033928880414 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=21)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, out
** Event #36 t=0.033928880414 WirelessB.hostA.udp (Udp, id=51) on UDPData-2 (inet::Packet, id=186)
INFO: Sending app packet UDPData-2 over ipv4.
INFO (MessageDispatcher)WirelessB.hostA.tn: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipOut, outGate
INFO (MessageDispatcher)WirelessB.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO (MessageDispatcher)WirelessB.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #37 t=0.033928880414 WirelessB.hostA.ipv4.ip (IPv4, id=75) on UDPData-2 (inet::Packet, id=186)
INFO: Received (inet::Packet)UDPData-2 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
DETAIL: Sending datagram 'UDPData-2' with destination = 10.0.0.2
INFO: Routing (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B with destination = 10.0.0.2, output interface = wlan0, next hop address = <unspec>
INFO: Sending (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B to output interface = wlan0.
INFO (MessageDispatcher)WirelessB.hostA.nl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← ip.queueOut, outGate = (omnetpp::cGate)in[0]
INFO (MessageDispatcher)WirelessB.hostA.cb: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← nl.out[1], outGate = (omnetpp::cGate)out[
INFO (MessageDispatcher)WirelessB.hostA.bl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← cb.out[0], outGate = (omnetpp::cGate)out[
INFO (MessageDispatcher)WirelessB.hostA.li: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← bl.out[1], outGate = (omnetpp::cGate)out[
** Event #38 t=0.033928880414 WirelessB.hostA.wlan[0].queue (DropTailQueue, id=59) on UDPData-2 (inet::Packet, id=186)
INFO: Pushing packet, packet = (Packet)UDPData-2 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, totalLengthField = 1028 B, identification = 2, i
** Event #39 t=0.039323230943 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=61) on selfmsg transmissionTimer (omnetpp::cMessage, id=10)
INFO: Transmission ended: (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) lengt
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
INFO (DropTailQueue)WirelessB.hostA.wlan[0].queue: Pulling packet, packet = (Packet)UDPData-2 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, tot
INFO (AckingMac)WirelessB.hostA.wlan[0].mac: Received (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B for transmission
INFO (AckingMac)WirelessB.hostA.wlan[0].mac: Starting transmission of (inet::Packet)UDPData-2 (1051 B) (inet::SequenceChunk) length = 1051 B
INFO: Radio mode changed from RECEIVER to TRANSMITTER.
INFO: Changing radio reception state from IDLE to UNDEFINED.
INFO: Changing radio transmission state from UNDEFINED to IDLE.
** Event #40 t=0.039323250545 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=61) on UDPData-2 (inet::Packet, id=187)
INFO: Transmission started: (inet::physicalLayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B WHOLE as
INFO: Changing radio transmission state from IDLE to TRANSMITTING.
INFO: Changing radio transmitted signal part from NONE to WHOLE.
** Event #41 t=0.039323784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on selfmsg receptionTimer (omnetpp::cMessage, id=176)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 106
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #42 t=0.039323784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on UDPData-2 (inet::physicalLayer::WirelessSignal, id=193)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #43 t=0.039323784508 WirelessB.hostR2.wlan[0].mac (AckingMac, id=181) on UDPData-1 (inet::Packet, id=202)
INFO: Frame 'UDPData-1' not destined to us, discarding
** Event #44 t=0.039323922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on selfmsg receptionTimer (omnetpp::cMessage, id=179)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 106
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #45 t=0.039323922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on UDPData-2 (inet::physicalLayer::WirelessSignal, id=191)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #46 t=0.039323922865 WirelessB.hostR1.wlan[0].mac (AckingMac, id=141) on UDPData-1 (inet::Packet, id=205)
INFO: Frame 'UDPData-1' not destined to us, discarding
** Event #47 t=0.039324334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on selfmsg receptionTimer (omnetpp::cMessage, id=182)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 106
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #48 t=0.039324334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on UDPData-2 (inet::physicalLayer::WirelessSignal, id=195)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #49 t=0.039324334628 WirelessB.hostR3.wlan[0].mac (AckingMac, id=221) on UDPData-1 (inet::Packet, id=208)
INFO: Frame 'UDPData-1' not destined to us, discarding
** Event #50 t=0.039324584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on selfmsg receptionTimer (omnetpp::cMessage, id=185)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 106
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.

```

Step 5 Taking Interference into Account

The radio model is upgraded from UnitDiskRadio (ideal, binary reception) to APSK radio with a scalar analog model. This introduces SINR (Signal-to-Interference-plus-Noise Ratio) based reception decisions, meaning transmissions from other nodes now cause real interference. The IdealMedium is replaced by RadioMedium with a path loss model. You will observe that some packets are lost due to interference, which was not possible in the previous ideal model.



```
** Event #30 t=0.030820584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on UDPData-1 (inet::physicallayer::WirelessSignal, id=384)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #31 t=0.030820584801 WirelessB.hostB.wlan[0].mac (AckingMac, id=181) on UDPData-0 (inet::Packet, id=406)
INFO: Passing up contained packet 'UDPData-0' to higher layer
INFO (MessageDispatcher)WirelessB.hostB.l1: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← l1.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessB.hostB.ipv4.lip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.ifir
** Event #32 t=0.030820584801 WirelessB.hostB.ipv4.lip (Ipv4, id=116) on UDPData-0 (inet::Packet, id=406)
INFO: Received (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram '' with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-0 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessB.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessB.hostB.ipv4.up: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessB.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #33 t=0.030820584801 WirelessB.hostB.udp (Udp, id=92) on UDPData-0 (inet::Packet, id=406)
INFO: Packet UDPData-0 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessB.hostB.at: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appOut, outGate = (omnetpp::cGate)out[0] →
** Event #34 t=0.030820584801 WirelessB.hostB.app[0] (UdpSink, id=95) on UDPData-0 (inet::Packet, id=406)
INFO: Received packet: (inet::Packet)UDPData-0 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
** Event #35 t=0.033928880414 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=243)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, out
```

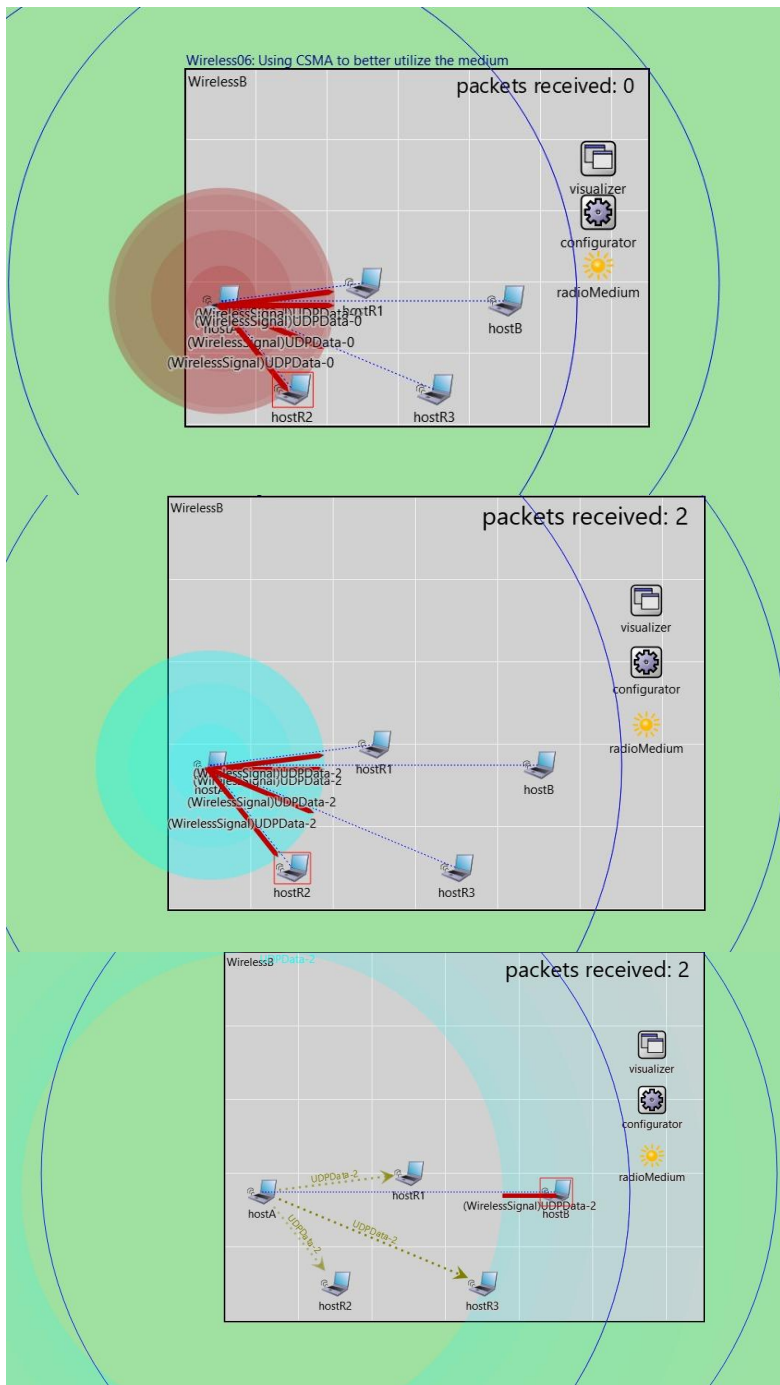
```
** Event #40 t=0.0393232590545 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=61) on UDPData-2 (inet::Packet, id=409)
INFO: Transmission started: (inet::physicallayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio transmission state from TDLE to TRANSMITTING.
INFO: Changing radio transmitted signal part from NONE to WHOLE.
** Event #41 t=0.039323784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on selfmsg receptionTimer (omnetpp::cMessage, id=398)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #42 t=0.039323784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=182) on UDPData-2 (inet::physicallayer::WirelessSignal, id=415)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #43 t=0.039323784508 WirelessB.hostR2.wlan[0].mac (AckingMac, id=181) on UDPData-1 (inet::Packet, id=424)
INFO: Frame 'UDPData-1' not destined to us, discarding
** Event #44 t=0.039323922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on selfmsg receptionTimer (omnetpp::cMessage, id=401)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #45 t=0.039323922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=142) on UDPData-2 (inet::physicallayer::WirelessSignal, id=413)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #46 t=0.039323922865 WirelessB.hostR1.wlan[0].mac (AckingMac, id=141) on UDPData-1 (inet::Packet, id=427)
INFO: Frame 'UDPData-1' not destined to us, discarding
** Event #47 t=0.039324334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on selfmsg receptionTimer (omnetpp::cMessage, id=404)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #48 t=0.039324334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=222) on UDPData-2 (inet::physicallayer::WirelessSignal, id=417)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #49 t=0.039324334628 WirelessB.hostR3.wlan[0].mac (AckingMac, id=221) on UDPData-1 (inet::Packet, id=430)
INFO: Frame 'UDPData-1' not destined to us, discarding
```

```
** Event #50 t=0.039324584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on selfmsg receptionTimer (omnetpp::cMessage, id=407)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.504 ms 1063 B) (inet::Packet)UDPData-1 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Sending up (inet::Packet)UDPData-1 (1051 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from RECEIVING to BUSY.
** Event #51 t=0.039324584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=102) on UDPData-2 (inet::physicallayer::WirelessSignal, id=411)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-2 (8.504 ms 1063 B) (inet::Packet)UDPData-2 (1063 B) (inet::SequenceChunk) length = 1063 B
INFO: Changing radio reception state from BUSY to RECEIVING.
** Event #52 t=0.039324584801 WirelessB.hostB.wlan[0].mac (AckingMac, id=101) on UDPData-1 (inet::Packet, id=433)
INFO: Passing up contained packet 'UDPData-1' to higher layer
INFO (MessageDispatcher)WirelessB.hostB.l1: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← l1.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessB.hostB.ipv4.lip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.ifir
** Event #53 t=0.039324584801 WirelessB.hostB.ipv4.lip (Ipv4, id=116) on UDPData-1 (inet::Packet, id=433)
INFO: Received (inet::Packet)UDPData-1 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram '' with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-1 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessB.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessB.hostB.ipv4.up: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessB.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #54 t=0.039324584801 WirelessB.hostB.udp (Udp, id=92) on UDPData-1 (inet::Packet, id=433)
INFO: Packet UDPData-1 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessB.hostB.at: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appOut, outGate = (omnetpp::cGate)out[0] →
** Event #55 t=0.039324584801 WirelessB.hostB.app[0] (UdpSink, id=95) on UDPData-1 (inet::Packet, id=433)
INFO: Received packet: (inet::Packet)UDPData-1 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
```

```
** Event #63 t=0.047828584801 WirelessB.hostB.wlan[0].mac (AckingMac, id=101) on UDPData-2 (inet::Packet, id=442)
INFO: Passing up contained packet 'UDPData-2' to higher layer
INFO (MessageDispatcher)WirelessB.hostB.l1: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← l1.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessB.hostB.ipv4.lip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.ifir
** Event #66 t=0.047828584801 WirelessB.hostB.ipv4.lip (Ipv4, id=116) on UDPData-2 (inet::Packet, id=442)
INFO: Received (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1063 B from network.
DETAIL: Received datagram '' with dest=10.0.0.2
INFO: Delivering (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1063 B locally.
INFO: Passing up to protocol udp(75)
INFO (MessageDispatcher)WirelessB.hostB.ipv4.mp: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[3] ← ip.transportOut
INFO (MessageDispatcher)WirelessB.hostB.ipv4.up: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← mp.out[0], out
INFO (MessageDispatcher)WirelessB.hostB.tn: Dispatching packet to protocol, protocol = udp(75), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← ipv4.transportOut,
** Event #67 t=0.047828584801 WirelessB.hostB.udp (Udp, id=92) on UDPData-2 (inet::Packet, id=442)
INFO: Packet UDPData-2 received from network, dest port 5000
INFO: Sending payload up to socket sockId=1
INFO (MessageDispatcher)WirelessB.hostB.at: Dispatching packet to socket, socketId = 1, inGate = (omnetpp::cGate)in[1] ← udp.appOut, outGate = (omnetpp::cGate)out[0] →
** Event #68 t=0.047828584801 WirelessB.hostB.app[0] (UdpSink, id=95) on UDPData-2 (inet::Packet, id=442)
INFO: Received packet: (inet::Packet)UDPData-2 (1000 B) (inet::SequenceChunk) length = 1063 B (1000 bytes) 10.0.0.1:1025 → 10.0.0.2:5000 TTL=32 TOS=0 DSCP=0 on ifID=101
** Event #69 t=0.049324584801 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=222)
** Event #70 t=0.053995414818 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=243)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, out
```

Step 6 Using CSMA to Better Utilize the Medium

A proper CSMA (Carrier Sense Multiple Access) MAC protocol is introduced, replacing the previous MAC-less setup. Nodes now listen to the channel before transmitting and back off if the medium is busy. This is the standard technique used in real-world 802.11 Wi-Fi. With CSMA enabled, collisions are reduced, channel utilisation improves, and throughput increases compared to Step 5. The NED change involves setting the wlan[*].mac.type to CsmCaMac.



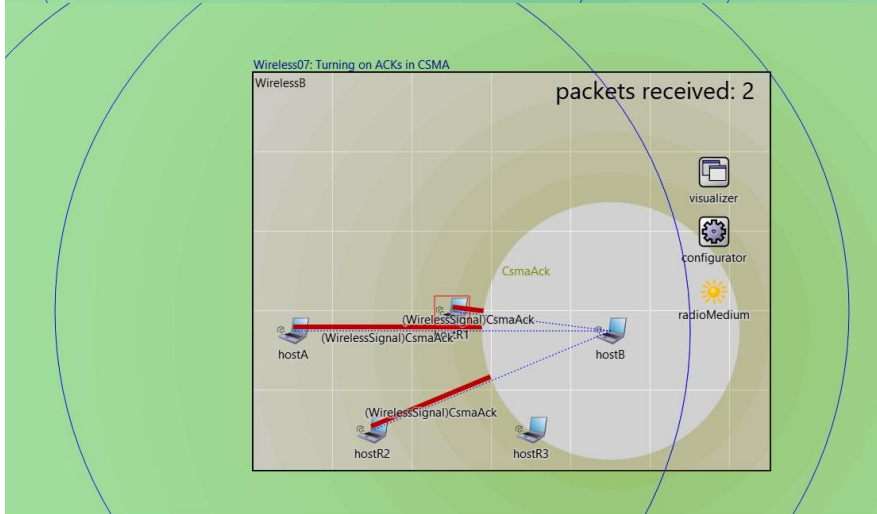
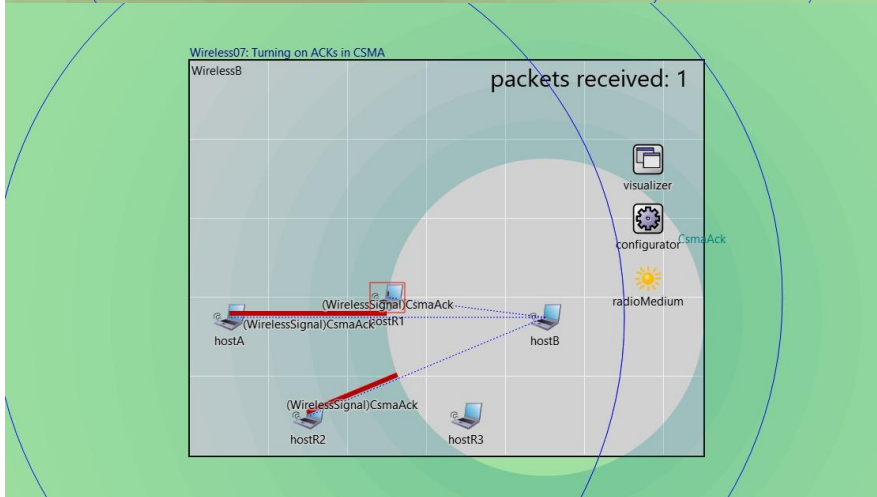
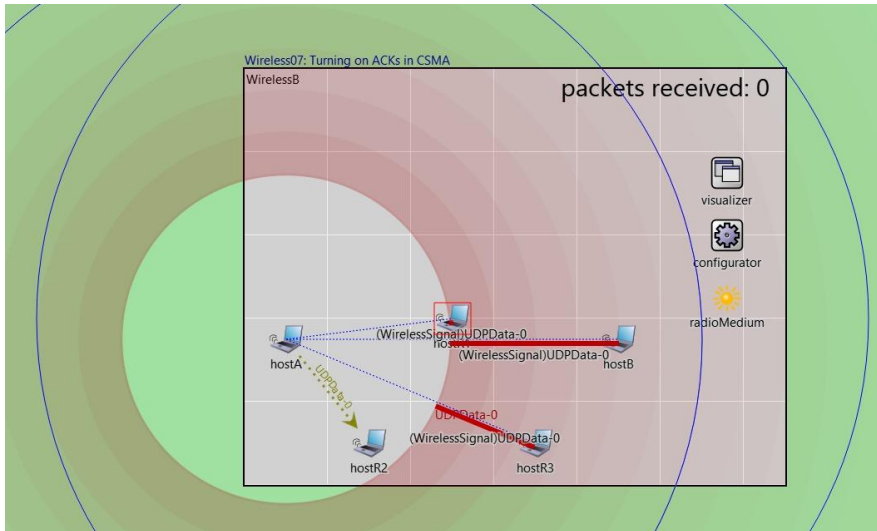

```

** Event #65 t=0.053978277646 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=62) on selfmsg transmissionTimer (omnetpp::cMessage, id=16)
INFO: Transmission ended: (inet::physicallayer::WirelessSignal)UDPData-2 (8.536 ms 1067 B) (inet::Packet)UDPData-2 (1067 B) (inet::SequenceChunk) length = 1067 B WHOLE as L
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: (DropTailQueue)WirelessB.hostA.wlan[0].queue: Pulling packet, packet = (Packet)UDPData-3 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, tot
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: frame (inet::Packet)UDPData-3 (1028 B) (inet::SequenceChunk) length = 1028 B received from higher layer, receiver = 0A-AA-00-00-00-00
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: scheduling backoff timer
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: generating backoff slot number for retry: 0
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: generated backoff slot number: 25 , cw: 31
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: backoff period set to 0.0005
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
** Event #66 t=0.053978811609 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=186) on selfmsg receptionTimer (omnetpp::cMessage, id=239)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-2 (8.536 ms 1067 B) (inet::Packet)UDPData-2 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-2 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #67 t=0.053978811609 WirelessB.hostR2.wlan[0].mac (CsmaCaMac, id=185) on UDPData-2 (inet::Packet, id=245)
INFO: received message from lower layer: (inet::Packet)UDPData-2 (1055 B) (inet::SequenceChunk) length = 1067 B
** Event #68 t=0.053978949966 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=145) on selfmsg receptionTimer (omnetpp::cMessage, id=240)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-2 (8.536 ms 1067 B) (inet::Packet)UDPData-2 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-2 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #69 t=0.053978949966 WirelessB.hostR1.wlan[0].mac (CsmaCaMac, id=144) on UDPData-2 (inet::Packet, id=247)
INFO: received message from lower layer: (inet::Packet)UDPData-2 (1055 B) (inet::SequenceChunk) length = 1067 B
** Event #70 t=0.053979361729 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=227) on selfmsg receptionTimer (omnetpp::cMessage, id=241)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-2 (8.536 ms 1067 B) (inet::Packet)UDPData-2 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-2 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #60 t=0.050048584801 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=0)
** Event #61 t=0.052808868453 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=27)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, ou
** Event #62 t=0.052808868453 WirelessB.hostA.udp (Udp, id=51) on UDPData-3 (inet::Packet, id=243)
INFO: Sending app packet UDPData-3 over ipv4.
INFO (MessageDispatcher)WirelessB.hostA.tn: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.ipv4.udp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO (MessageDispatcher)WirelessB.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #63 t=0.052808868453 WirelessB.hostA.ipv4.ip (Ipv4, id=76) on UDPData-3 (inet::Packet, id=243)
INFO: Received (inet::Packet)UDPData-3 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
DETAIL: Sending datagram 'UDPData-3' with destination = 10.0.0.2
INFO: Routing (inet::Packet)UDPData-3 (1028 B) (inet::SequenceChunk) length = 1028 B with destination = 10.0.0.2, output interface = wlan0, next hop address = <unspec>
INFO: Sending (inet::Packet)UDPData-3 (1028 B) (inet::SequenceChunk) length = 1028 B to output interface = wlan0.
INFO (MessageDispatcher)WirelessB.hostA.ipv4.ip: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← ip.queueOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.nl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← ipv4.ifOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.cb: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← nl.out[1], outGate = (omnetpp::cGate)out[1]
INFO (MessageDispatcher)WirelessB.hostA.bl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← cb.out[0], outGate = (omnetpp::cGate)out[0]
INFO (MessageDispatcher)WirelessB.hostA.li: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← bl.out[1], outGate = (omnetpp::cGate)out[1]
** Event #64 t=0.052808868453 WirelessB.hostA.wlan[0].queue (DropTailQueue, id=60) on UDPData-3 (inet::Packet, id=243)
INFO: Pushing packet, packet = (Packet)UDPData-3 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, totalLengthField = 1028 B, identification = 3, i
** Event #35 t=0.031512334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=227) on UDPData-1 (inet::physicallayer::WirelessSignal, id=206)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from IDLE to RECEIVING.
** Event #36 t=0.031512584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=104) on UDPData-1 (inet::physicallayer::WirelessSignal, id=200)
INFO: Reception started: attempting (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from IDLE to RECEIVING.
** Event #37 t=0.040047282865 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=62) on selfmsg transmissionTimer (omnetpp::cMessage, id=16)
INFO: Transmission ended: (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B WHOLE as L
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
** Event #38 t=0.040047784508 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=186) on selfmsg receptionTimer (omnetpp::cMessage, id=212)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #39 t=0.040047784508 WirelessB.hostR2.wlan[0].mac (CsmaCaMac, id=185) on UDPData-1 (inet::Packet, id=217)
INFO: received message from lower layer: (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
** Event #40 t=0.040047922865 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=145) on selfmsg receptionTimer (omnetpp::cMessage, id=213)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #41 t=0.040047922865 WirelessB.hostR1.wlan[0].mac (CsmaCaMac, id=144) on UDPData-1 (inet::Packet, id=219)
INFO: received message from lower layer: (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
** Event #42 t=0.040048334628 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=227) on selfmsg receptionTimer (omnetpp::cMessage, id=214)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #43 t=0.040048334628 WirelessB.hostR3.wlan[0].mac (CsmaCaMac, id=226) on UDPData-1 (inet::Packet, id=221)
INFO: received message from lower layer: (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
** Event #44 t=0.040048584801 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=104) on selfmsg receptionTimer (omnetpp::cMessage, id=215)
INFO: Reception ended: successfully for (inet::physicallayer::WirelessSignal)UDPData-1 (8.536 ms 1067 B) (inet::Packet)UDPData-1 (1067 B) (inet::SequenceChunk) length = 1067 B
INFO: Sending up (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #45 t=0.040048584801 WirelessB.hostB.wlan[0].mac (CsmaCaMac, id=103) on UDPData-1 (inet::Packet, id=223)
INFO: received message from lower layer: (inet::Packet)UDPData-1 (1055 B) (inet::SequenceChunk) length = 1067 B
INFO (MessageDispatcher)WirelessB.hostB.li: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← wlan[0].upperLayer
INFO (MessageDispatcher)WirelessB.hostB.bl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← li.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.cb: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[0] ← bl.out[0], outGate
INFO (MessageDispatcher)WirelessB.hostB.nl: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[1] ← cb.out[1], outGate
INFO (MessageDispatcher)WirelessB.hostB.ipv4.ip: Dispatching packet to protocol, protocol = ipv4(50), servicePrimitive = 2, inGate = (omnetpp::cGate)in[2] ← <parent>.ifir

```

Step 7 Turning On ACKs in CSMA

Acknowledgements (ACKs) are enabled in the CSMA MAC layer. Without ACKs, a sender has no way to know if a packet was received. With ACKs enabled (useAck = true), lost packets can be detected and retransmitted at the MAC layer. This introduces retransmission overhead but greatly improves reliability. You will observe that end-to-end packet delivery ratio increases while raw throughput may slightly decrease due to ACK traffic.



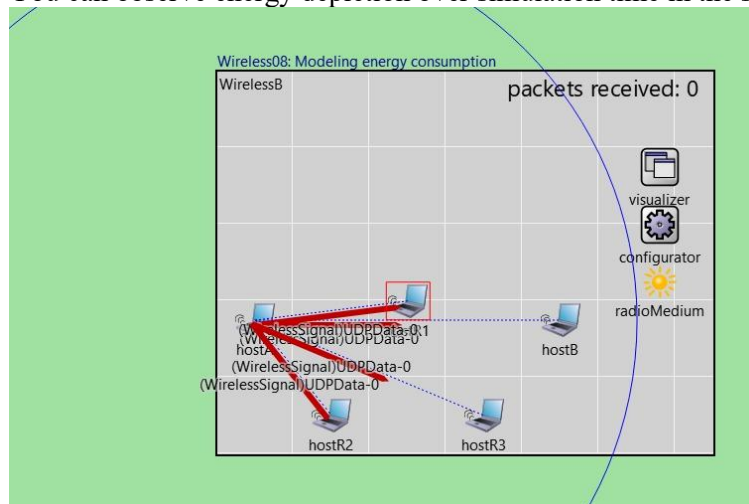

```

** Event #80 t=0.040652587569 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=61) on CsmaAck (inet::Packet, id=527)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: canceling ACK timer
** Event #81 t=0.041422584801 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=250)
** Event #82 t=0.041673919057 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=250)
** Event #83 t=0.044962277646 WirelessB.hostA.app[0] (UdpBasicApp, id=54) on selfmsg sendTimer (inet::ClockEvent, id=277)
INFO (MessageDispatcher)WirelessB.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, ou
** Event #84 t=0.044962277646 WirelessB.hostA.udp (Udp, id=51) on UDPData-2 (inet::Packet, id=528)
INFO: Sending app packet UDPData-2 over ipv4.
INFO (MessageDispatcher)WirelessB.hostA.tn: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipOut, outGate
INFO (MessageDispatcher)WirelessB.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO (MessageDispatcher)WirelessB.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #85 t=0.044962277646 WirelessB.hostA.ipv4.ip (Ipv4, id=76) on UDPData-2 (inet::Packet, id=528)
INFO: Received (inet::Packet)UDPData-2 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
DETAIL: Sending datagram 'UDPData-2' with destination = 10.0.0.2
INFO: Routing (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B with destination = 10.0.0.2, output interface = wlan0, next hop address = <unspec>
INFO: Sending (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B to output interface = wlan0.
INFO (MessageDispatcher)WirelessB.hostA.ipv4.lp: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← ip.queueOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.nl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← ipv4.ifOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.cb: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← nl.out[1], outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.bl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← cb.out[0], outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.l1: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← bl.out[1], outGate = (omnetpp::cGate)out
** Event #70 t=0.040412337396 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=186) on CsmaAck (inet::physicalLayer::WirelessSignal, id=508)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as Un
INFO: Changing radio reception state from IDLE to RECEIVING.
** Event #71 t=0.040412587569 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=62) on CsmaAck (inet::physicalLayer::WirelessSignal, id=504)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as Un
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio received signal part from NONE to WHOLE.
** Event #72 t=0.040651253313 WirelessB.hostB.wlan[0].radio (UnitDiskRadio, id=104) on selfmsg transmissionTimer (omnetpp::cMessage, id=293)
INFO: Transmission ended: (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskTrans
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
** Event #73 t=0.040651787276 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=227) on selfmsg receptionTimer (omnetpp::cMessage, id=516)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as
INFO: Sending up (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #74 t=0.040651787276 WirelessB.hostR3.wlan[0].mac (CsmaCaMac, id=226) on CsmaAck (inet::Packet, id=521)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
** Event #75 t=0.040651925633 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=145) on selfmsg receptionTimer (omnetpp::cMessage, id=517)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as
INFO: Sending up (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #76 t=0.040651925633 WirelessB.hostR1.wlan[0].mac (CsmaCaMac, id=144) on CsmaAck (inet::Packet, id=523)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
** Event #77 t=0.040652337396 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=186) on selfmsg receptionTimer (omnetpp::cMessage, id=518)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as
INFO: Sending up (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #78 t=0.040652337396 WirelessB.hostR2.wlan[0].mac (CsmaCaMac, id=185) on CsmaAck (inet::Packet, id=525)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
** Event #79 t=0.040652587569 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=62) on selfmsg receptionTimer (omnetpp::cMessage, id=519)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as
INFO: Sending up (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.

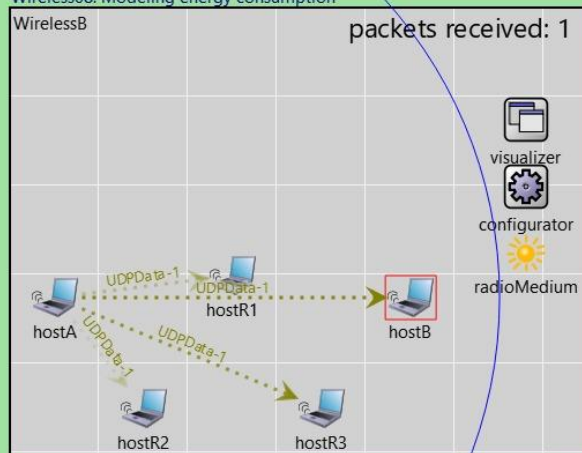
```

Step 8 Modeling Energy Consumption

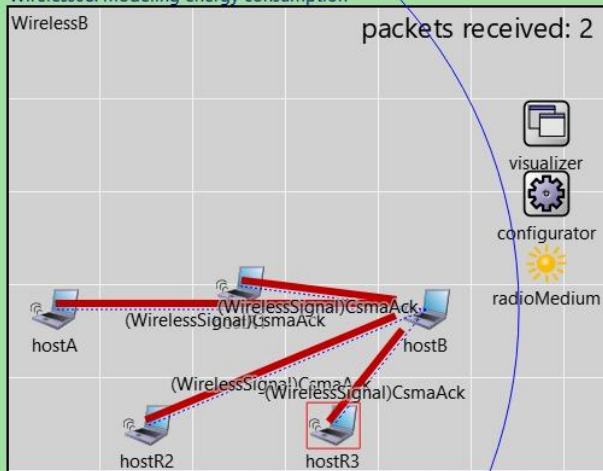
An energy storage and consumption model is added to each node. The wlan radio interface is configured to report power draw in each state (transmitting, receiving, idle, sleep). A SimpleEpEnergyStorage module tracks the remaining energy of each node. This step does not change communication behaviour but adds an important real-world constraint: nodes have a finite battery. You can observe energy depletion over simulation time in the scalar results.



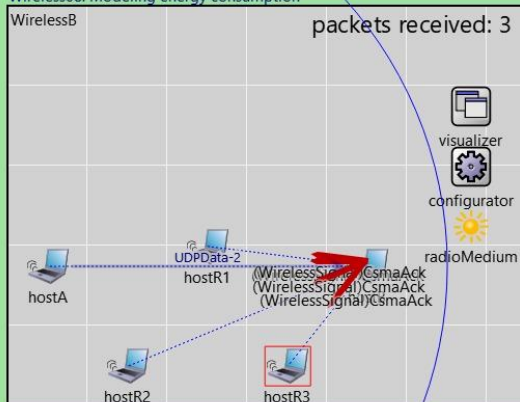
Wireless08: Modeling energy consumption



Wireless08: Modeling energy consumption



Wireless08: Modeling energy consumption



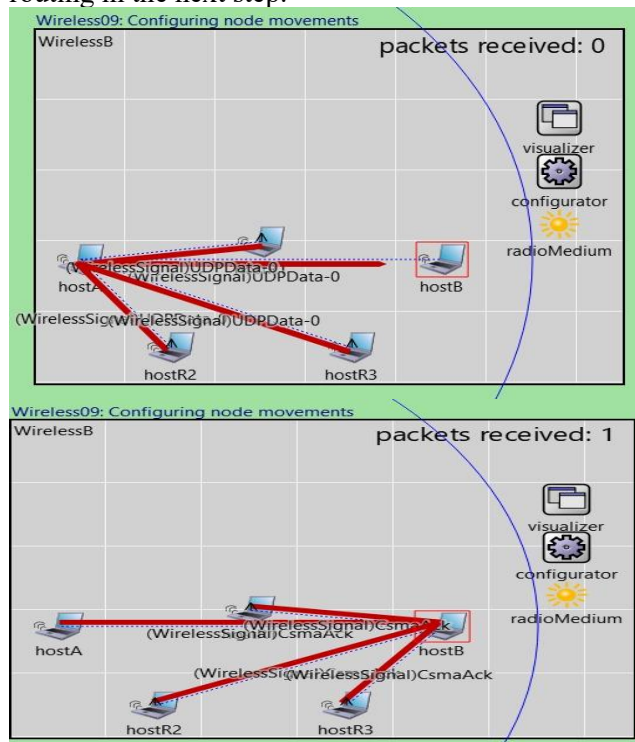
```

* Event #70 t=0.040412337396 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=193) on Csmack (inet::physicalLayer:WirelessSignal, id=801)
INFO: Reception started: attempting (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Changing radio reception state from IDLE to RECEIVING.
* Event #71 t=0.040412587569 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=63) on Csmack (inet::physicalLayer:WirelessSignal, id=797)
INFO: Reception started: attempting (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio received signal part from NONE to WHOLE.
* Event #72 t=0.040651233313 WirelessB.wlan[0].radio (UnitDiskRadio, id=107) on selfmsg transmissionTimer (omnetpp::cMessage, id=586)
INFO: Transmission ended: (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Changing radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
* Event #73 t=0.040651787276 WirelessB.hostR3.wlan[0].radio (UnitDiskRadio, id=236) on selfmsg receptionTimer (omnetpp::cMessage, id=809)
INFO: Reception ended: successfully for (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Sending up (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
* Event #74 t=0.040651787276 WirelessB.hostR3.wlan[0].mac (CsmaCaMac, id=235) on Csmack (inet::Packet, id=814)
INFO: received message from lower layer: (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
* Event #75 t=0.031673118764 WirelessB.hostR3.wlan[0].mac (CsmaCaMac, id=235) on Csmack (inet::Packet, id=761)
INFO: received message from lower layer: (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
* Event #76 t=0.031673257121 WirelessB.hostR1.wlan[0].radio (UnitDiskRadio, id=150) on selfmsg receptionTimer (omnetpp::cMessage, id=757)
INFO: Reception ended: successfully for (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Sending up (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
* Event #77 t=0.031673257121 WirelessB.hostR1.wlan[0].mac (CsmaCaMac, id=149) on Csmack (inet::Packet, id=763)
INFO: received message from lower layer: (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
* Event #78 t=0.031673608884 WirelessB.hostR2.wlan[0].radio (UnitDiskRadio, id=193) on selfmsg receptionTimer (omnetpp::cMessage, id=758)
INFO: Reception ended: successfully for (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Sending up (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
* Event #79 t=0.031673608884 WirelessB.hostR2.wlan[0].mac (CsmaCaMac, id=192) on Csmack (inet::Packet, id=765)
INFO: received message from lower layer: (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
* Event #80 t=0.031673919057 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=63) on selfmsg receptionTimer (omnetpp::cMessage, id=759)
INFO: Reception ended: successfully for (inet::physicalLayer:WirelessSignal)Csmack (240 us 30 B) (inet::Packet)Csmack (30 B) (inet::SequenceChunk) length = 30 B WHOLE as UnitDiskRadio
INFO: Sending up (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
* Event #81 t=0.031673919057 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on Csmack (inet::Packet, id=767)
INFO: received message from lower layer: (inet::Packet)Csmack (18 B) (inet::SequenceChunk) length = 30 B
INFO: canceling ACK timer
INFO: (omnetpp::cQueue)WirelessB.hostA.wlan[0].queue: Pulling packet, packet = (Packet)UDPData-1 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, totalLength = 1028 B]
INFO: frame (inet::Packet)UDPData-1 (1028 B) (inet::SequenceChunk) length = 1028 B received from higher layer, receiver = 0A-AA-00-00-00-02

```

Step 9 Configuring Node Movements

Mobility is introduced: R1 and R2 are given a LinearMobility model, causing them to move across the simulation area during the run. As nodes move, communication links may break or form dynamically. The static routes from Step 4 quickly become invalid as relay positions change. This step exposes a fundamental limitation of static routing in mobile networks and motivates the use of dynamic ad-hoc routing in the next step.



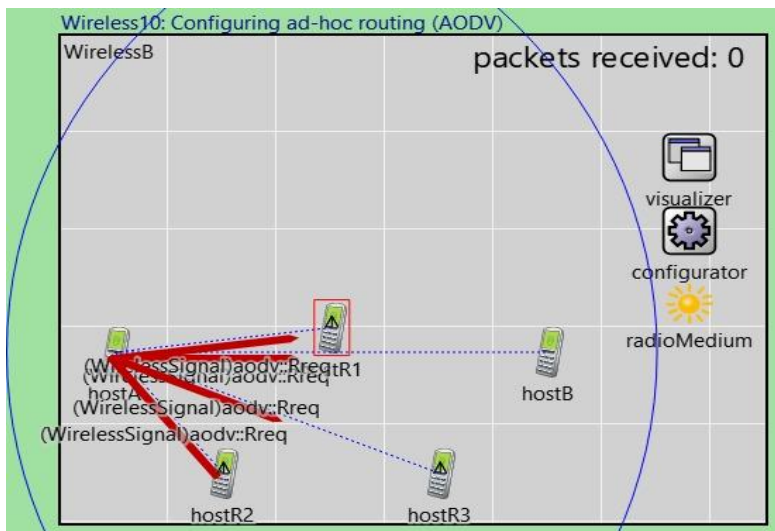
```

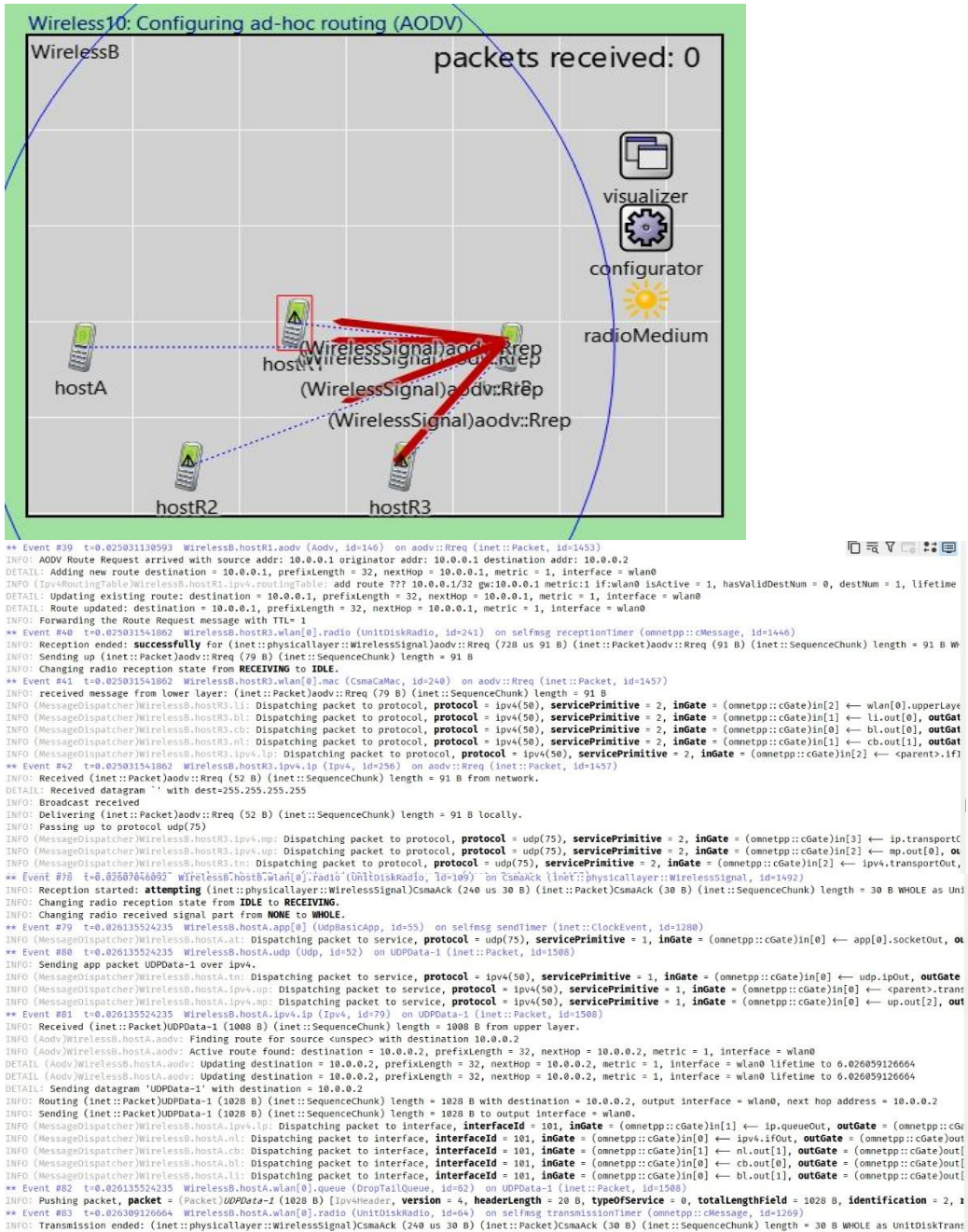
** Event #80 t=0.04062287309 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on CsmacAck (inet::Packet, id=1144)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: canceling ACK timer
** Event #81 t=0.041422584801 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=864)
** Event #82 t=0.041673919057 WirelessB.radioMedium (RadioMedium, id=4) on selfmsg removeNonInterferingTransmissions (omnetpp::cMessage, id=864)
** Event #83 t=0.044962277646 WirelessB.hostA.app[0] (UDPBasicApp, id=55) on selfmsg sendTimer (inet::ClockEvent, id=891)
INFO (MessageDispatcher)WirelessB.hostA.nl: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketOut, ou
** Event #84 t=0.044962277646 WirelessB.hostA.udp (udp, id=52) on UDPData-2 (inet::Packet, id=1145)
INFO: Sending app packet UDPData-2 over ipv4.
INFO (MessageDispatcher)WirelessB.hostA.tn: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipOut, outGate
INFO (MessageDispatcher)WirelessB.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO (MessageDispatcher)WirelessB.hostA.ipv4.no: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #85 t=0.044962277646 WirelessB.hostA.ipv4.ip (IPv4, id=76) on UDPData-2 (inet::Packet, id=1145)
INFO: Received (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B from upper layer.
DETAIL: Sending datagram 'UDPData-2' with destination = 10.0.0.2
INFO: Routing (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B with destination = 10.0.0.2, output interface = wlan0, next hop address = <unspec>
INFO: Sending (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B to output interface = wlan0.
INFO (MessageDispatcher)WirelessB.hostA.ipv4.ly: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← ip.queueOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.nl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← ipv4.ifOut, outGate = (omnetpp::cGate)out
INFO (MessageDispatcher)WirelessB.hostA.cb: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[1] ← nl.out[1], outGate = (omnetpp::cGate)out[
INFO (MessageDispatcher)WirelessB.hostA.bl: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← cb.out[0], outGate = (omnetpp::cGate)out[
INFO (MessageDispatcher)WirelessB.hostA.l1: Dispatching packet to interface, interfaceId = 101, inGate = (omnetpp::cGate)in[0] ← l1.out[1], outGate = (omnetpp::cGate)out[
** Event #86 t=0.044962277646 WirelessB.hostA.wlan[0].queue (DropTailQueue, id=81) on UDPData-2 (inet::Packet, id=1145)
INFO: Pushing packet, packet = (Packet)UDPData-2 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, totalLengthField = 1028 B, identification = 2, r
INFO: Pulling packet, packet = (Packet)UDPData-2 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, totalLengthField = 1028 B, identification = 2, r
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: frame (inet::Packet)UDPData-2 (1028 B) (inet::SequenceChunk) length = 1028 B received from higher layer, receiver = 0A-AA-00-00-00-02
INFO (CsmaCaMac)WirelessB.hostA.wlan[0].mac: scheduling DIFS timer
** Event #87 t=0.044962277646 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on selfmsg DIFS (omnetpp::cMessage, id=873)
** Event #45 t=0.031673919057 WirelessB.hostA.wlan[0].radio (UnitDiskRadio, id=63) on selfmsg receptionTimer (omnetpp::cMessage, id=1083)
INFO: Reception ended: successfully for (inet::PhysicalLayer)WirelessSignal)CsmaAck (240 us 30 B) (inet::Packet)CsmaAck (30 B) (inet::SequenceChunk) length = 30 B WHOLE as
INFO: Sending up (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #46 t=0.031673919057 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on CsmacAck (inet::Packet, id=1091)
INFO: received message from lower layer: (inet::Packet)CsmaAck (18 B) (inet::SequenceChunk) length = 30 B
INFO: canceling ACK timer
INFO (DropTailQueue)WirelessB.hostA.wlan[0].queue: Pulling packet, packet = (Packet)UDPData-1 (1028 B) [IPv4Header, version = 4, headerLength = 20 B, typeOfService = 0, tot
INFO: Frame (inet::Packet)UDPData-1 (1028 B) (inet::SequenceChunk) length = 1028 B received from higher layer, receiver = 0A-AA-00-00-00-02
INFO: scheduling DIFS timer
** Event #47 t=0.031723919057 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on selfmsg DIFS (omnetpp::cMessage, id=872)
INFO: received self message: (omnetpp::cMessage)DIFS
INFO: scheduling backoff timer
INFO: generating backoff slot number for retry: 0
INFO: generated backoff slot number: 7, cw: 31
INFO: backoff period set to 0.00014
** Event #48 t=0.031863919057 WirelessB.hostA.wlan[0].mac (CsmaCaMac, id=62) on selfmsg Backoff (omnetpp::cMessage, id=873)
INFO: received self message: (omnetpp::cMessage)Backoff
INFO: sending data frame UDPData-1
INFO (UnitDiskRadio)WirelessB.hostA.wlan[0].radio: Radio mode changed from RECEIVER to TRANSMITTER.
INFO (UnitDiskRadio)WirelessB.hostA.wlan[0].radio: Changing radio reception state from IDLE to UNDEFINED.
INFO (UnitDiskRadio)WirelessB.hostA.wlan[0].radio: Changing radio transmission state from UNDEFINED to IDLE.
INFO (UnitDiskRadio)WirelessB.hostA.wlan[0].radio: Changing radio received signal part from WHOLE to NONE.

```

Step 10 Configuring Ad-Hoc Routing (AODV)

AODV (Ad-hoc On-demand Distance Vector) routing is introduced to replace static routing. AODV discovers routes dynamically on demand: when hostA needs to reach hostB, it broadcasts a Route Request (RREQ); intermediate nodes relay it; hostB replies with a Route Reply (RREP). As nodes move and links break, AODV re-discovers routes automatically. This step shows how mobile ad-hoc networks (MANETs) maintain connectivity without any infrastructure.

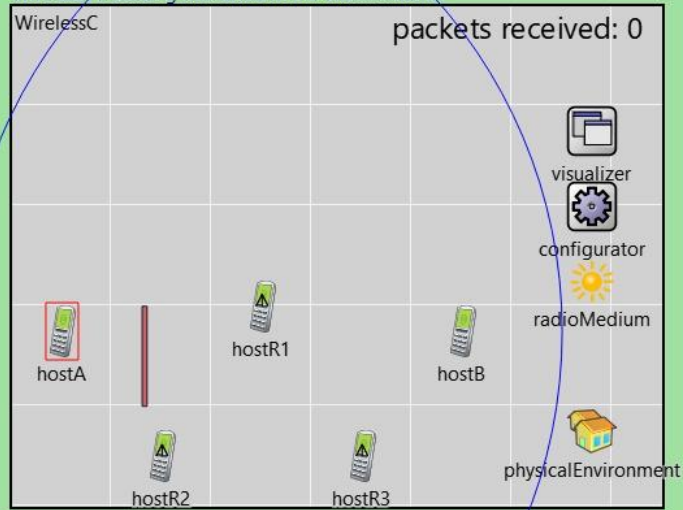




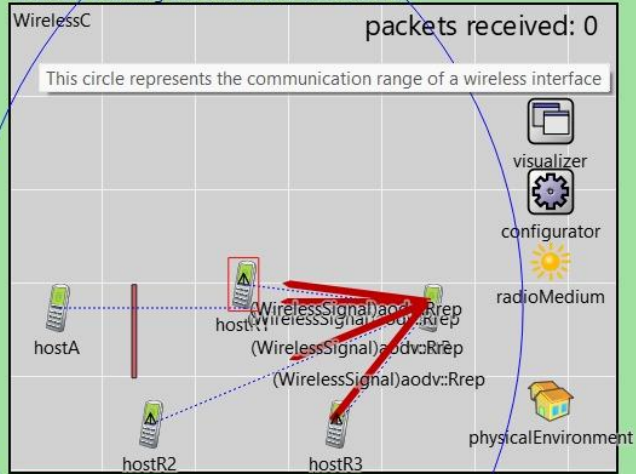
Step 11 Adding Obstacles to the Environment

A physical environment with obstacles (walls, buildings) is added using the PhysicalEnvironment module and an XML file describing obstacle geometry and material properties. Obstacles attenuate radio signals based on their material (e.g., concrete reduces signal strength more than wood). This step makes the propagation model much more realistic, as walls now block or degrade signals rather than signals propagating freely in all directions.

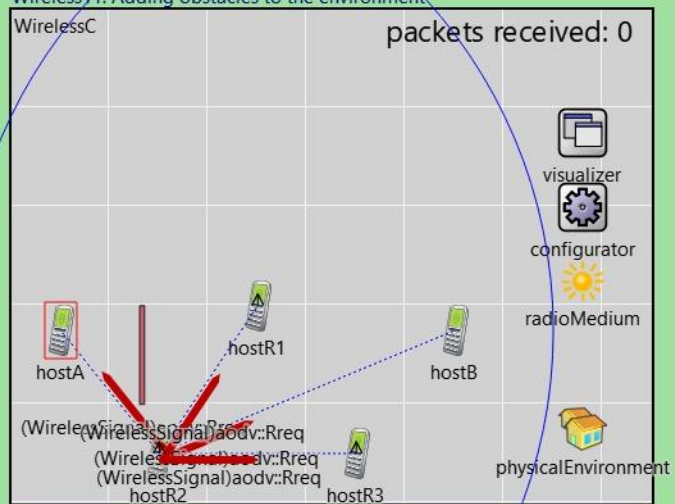
Wireless11: Adding obstacles to the environment

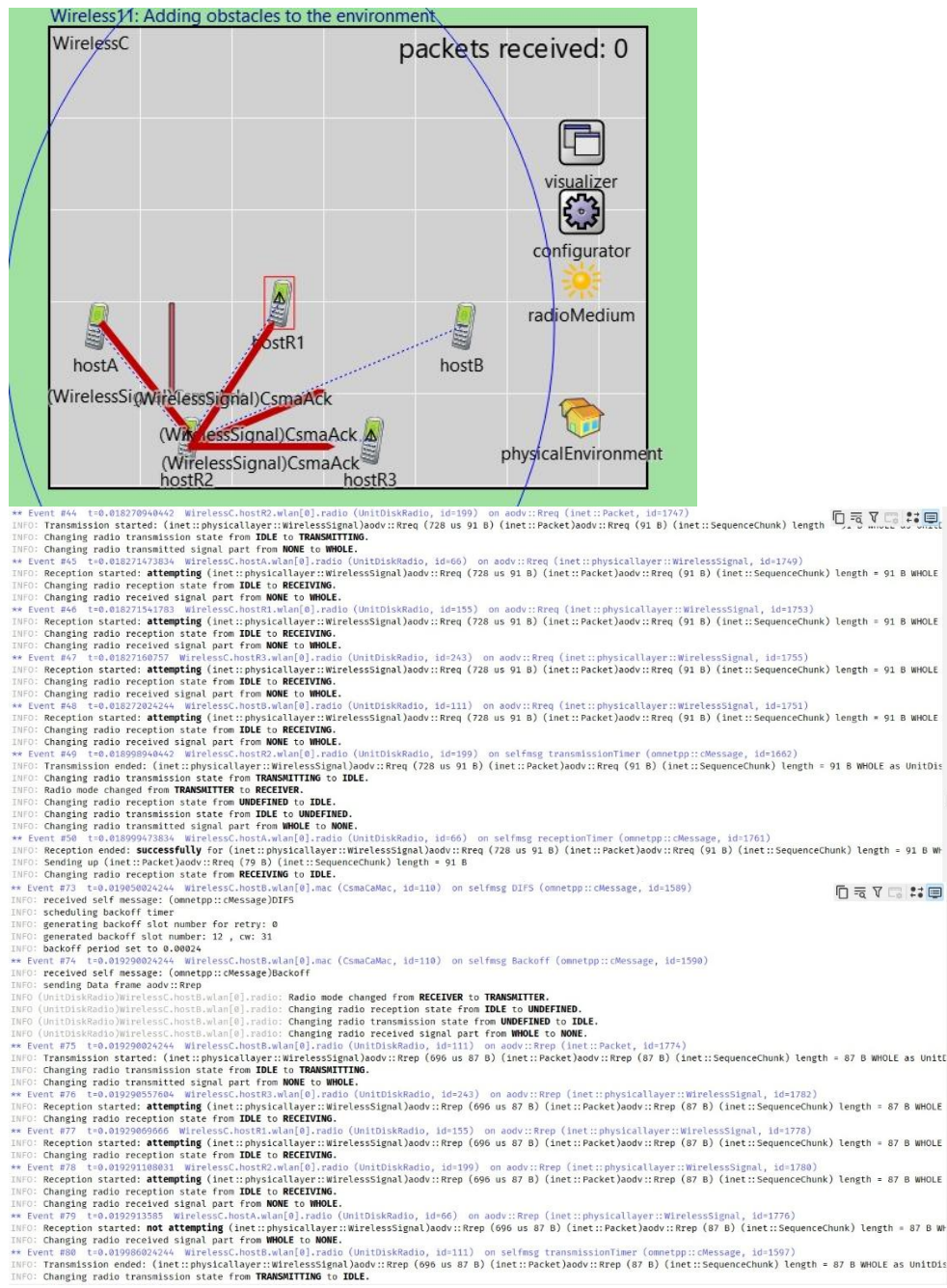


Wireless11: Adding obstacles to the environment



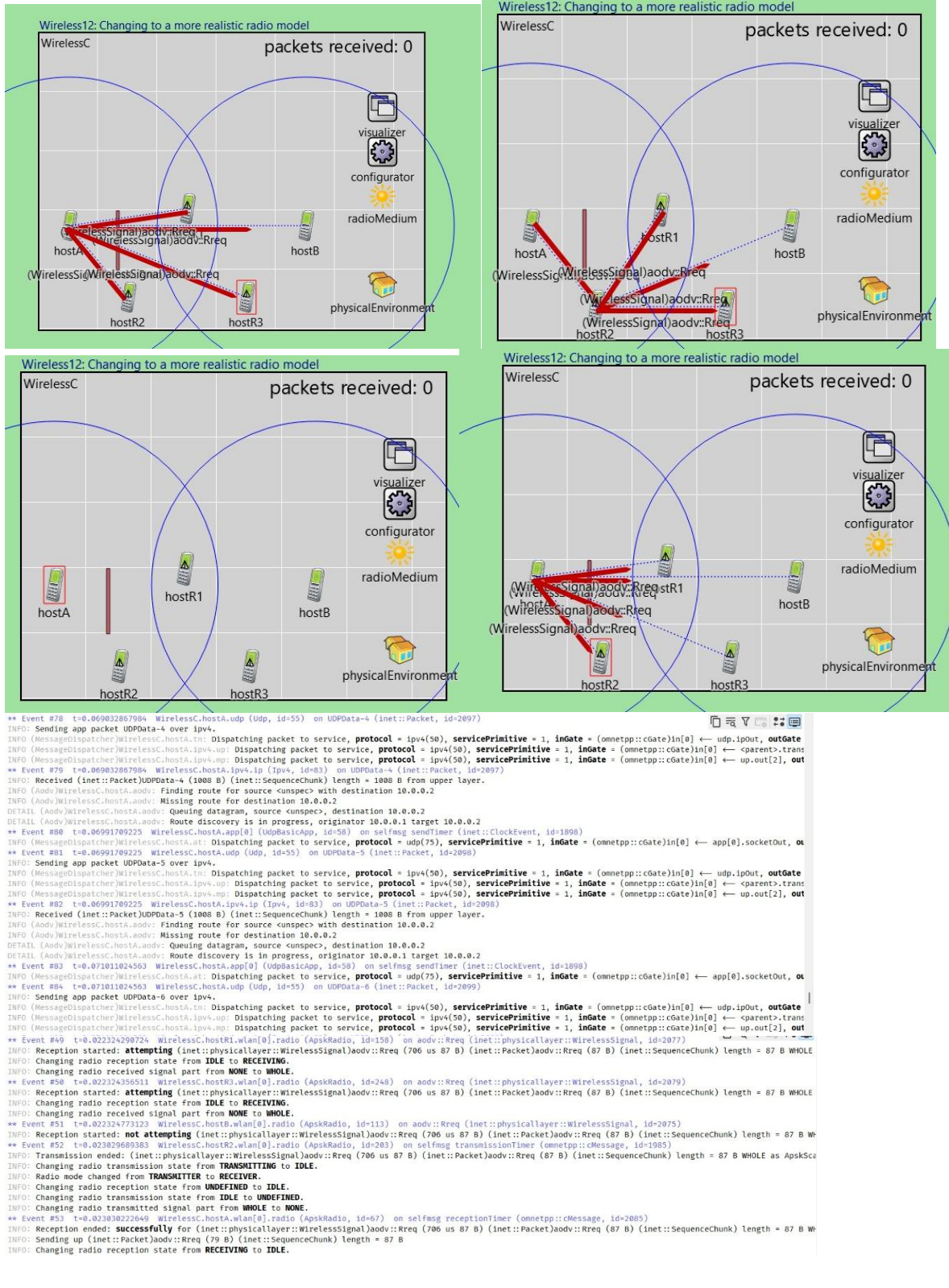
Wireless11: Adding obstacles to the environment





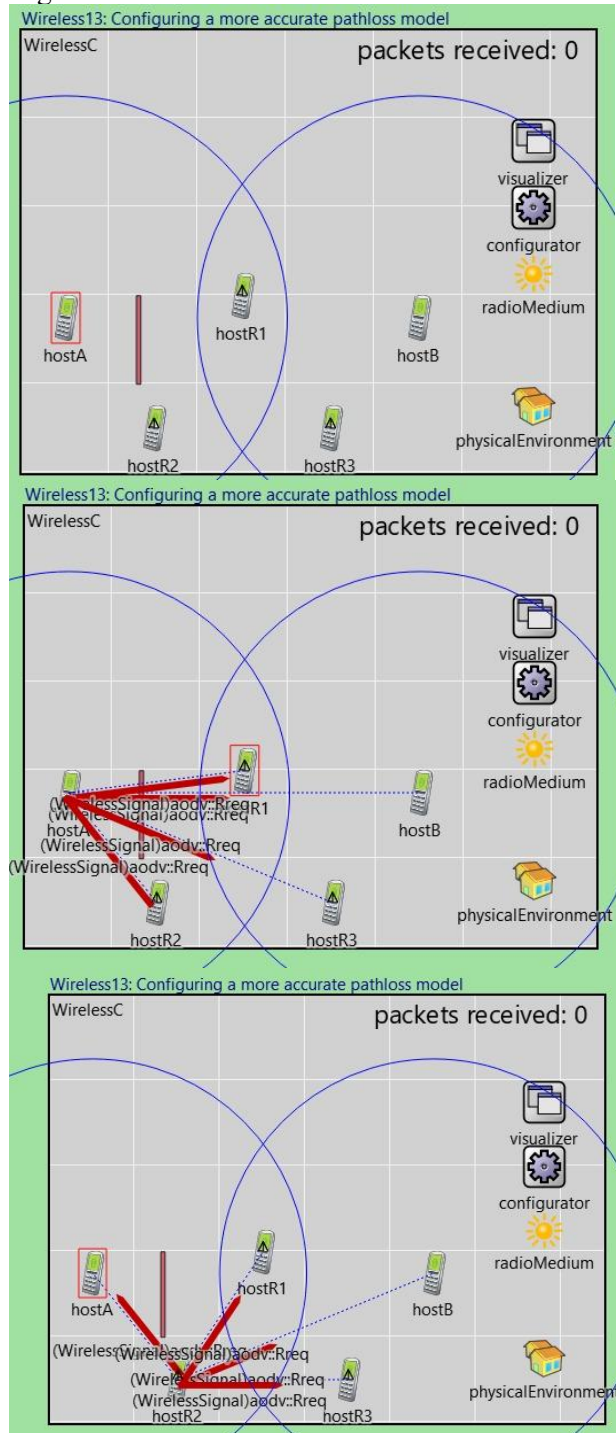
Step 12 Changing to a More Realistic Radio Model

The radio model is upgraded from the simplified scalar analog model to the dimensional analog model (Ieee80211DimensionalRadioMedium). This model represents signals in the time-frequency domain, allowing more accurate simulation of phenomena like spectral overlap and frequency-selective fading. The IEEE 802.11g standard parameters are used. This step bridges the gap between idealised simulation and realistic Wi-Fi behaviour, at the cost of higher computational complexity.



Step 13 Configuring a More Accurate Path Loss Model

The path loss model is refined by replacing the simple free-space formula with the TwoRayGroundReflection model. This model accounts for ground-reflected signals interfering with the direct path signal, which is a much more accurate representation of outdoor propagation. At short distances, the two-ray model behaves similarly to free-space, but at longer distances signal strength drops much faster. This affects the effective communication range and packet delivery at the network edges.



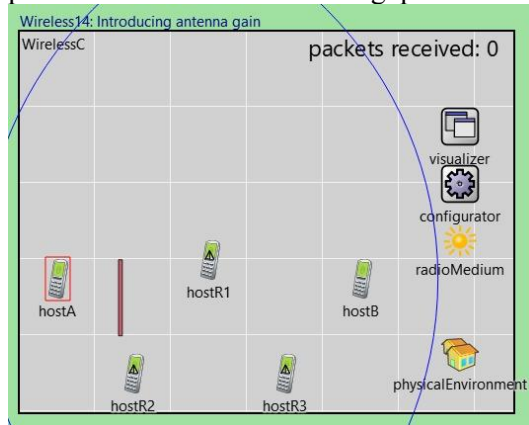
```

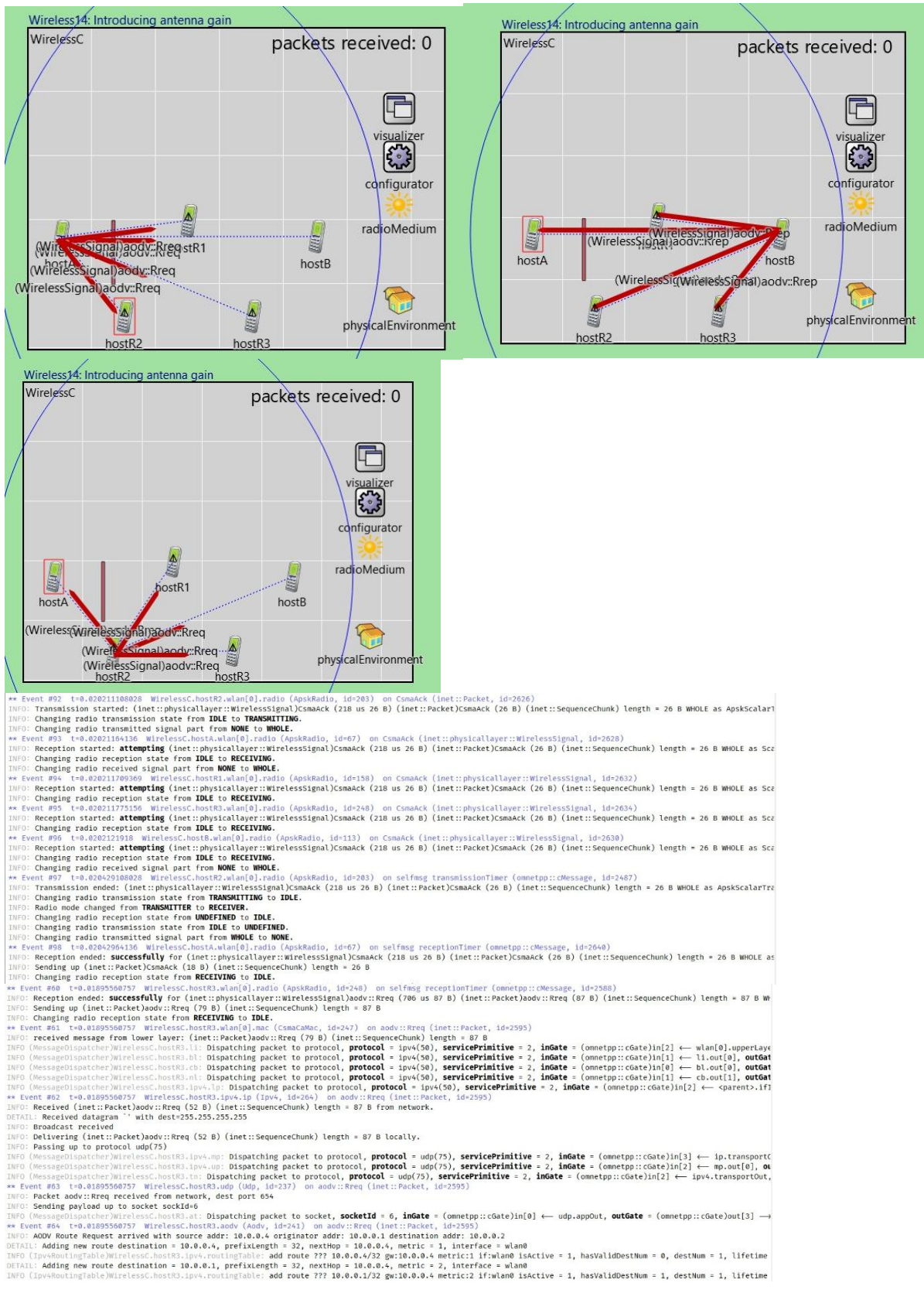
** Event #46 t=0.022323689383 WirelessC.hostR2.wlan[0].mac (CSmaMac, id=202) on selfmsg Backoff (omnetpp::cMessage, id=2244)
INFO: received self message: (omnetpp::cMessage)Backoff
INFO: sending data frame aodv::Rreq
INFO: (ApskRadio)WirelessC.hostR2.wlan[0].radio: Radio mode changed from RECEIVER to TRANSMITTER.
INFO: (ApskRadio)WirelessC.hostR2.wlan[0].radio: Changing radio reception state from IDLE to UNDEFINED.
INFO: (ApskRadio)WirelessC.hostR2.wlan[0].radio: Changing radio transmission state from UNDEFINED to IDLE.
INFO: (ApskRadio)WirelessC.hostR2.wlan[0].radio: Changing radio received signal part from WHOLE to NONE.
** Event #47 t=0.022323689383 WirelessC.hostR2.wlan[0].radio (ApskRadio, id=203) on aodv::Rreq (inet::Packet, id=2337)
INFO: Transmission started: (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE as ApskC
INFO: Changing radio transmission state from IDLE to TRANSMITTING.
INFO: Changing radio transmitted signal part from NONE to WHOLE.
** Event #48 t=0.022324022649 WirelessC.hostA.wlan[0].radio (ApskRadio, id=67) on aodv::Rreq (inet::physicalLayer::WirelessSignal, id=2339)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio received signal part from NONE to WHOLE.
** Event #49 t=0.022324297274 WirelessC.hostR1.wlan[0].radio (ApskRadio, id=158) on aodv::Rreq (inet::physicalLayer::WirelessSignal, id=2343)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio received signal part from NONE to WHOLE.
** Event #50 t=0.022324390211 WirelessC.hostR3.wlan[0].radio (ApskRadio, id=248) on aodv::Rreq (inet::physicalLayer::WirelessSignal, id=2343)
INFO: Reception started: attempting (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE
INFO: Changing radio reception state from IDLE to RECEIVING.
INFO: Changing radio received signal part from NONE to WHOLE.
** Event #51 t=0.022324773123 WirelessC.hostB.wlan[0].radio (ApskRadio, id=113) on aodv::Rreq (inet::physicalLayer::WirelessSignal, id=2341)
INFO: Reception started: not attempting (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE
** Event #52 t=0.023029689383 WirelessC.hostR2.wlan[0].radio (ApskRadio, id=203) on selfmsg transmissionTimer (omnetpp::cMessage, id=2251)
INFO: Transmission ended: (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE as ApskC
INFO: Radio transmission state from TRANSMITTING to IDLE.
INFO: Radio mode changed from TRANSMITTER to RECEIVER.
INFO: Changing radio reception state from UNDEFINED to IDLE.
INFO: Changing radio transmission state from IDLE to UNDEFINED.
INFO: Changing radio transmitted signal part from WHOLE to NONE.
** Event #53 t=0.023030222649 WirelessC.hostA.wlan[0].radio (ApskRadio, id=67) on selfmsg receptionTimer (omnetpp::cMessage, id=2351)
INFO: Reception ended: successfully for (inet::physicalLayer::WirelessSignal)aodv::Rreq (706 us 87 B) (inet::Packet)aodv::Rreq (87 B) (inet::SequenceChunk) length = 87 B WHOLE
INFO: Sending up (inet::Packet)aodv::Rreq (79 B) (inet::SequenceChunk) length = 87 B
INFO: Changing radio reception state from RECEIVING to IDLE.
** Event #77 t=0.069032867984 WirelessC.hostA.app[0] (UdpBasicApp, id=58) on selfmsg sendTimer (inet::ClockEvent, id=2164)
INFO: (MessageDispatcher)WirelessC.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketout, out
** Event #78 t=0.069032867984 WirelessC.hostA.udp (Udp, id=55) on UDPData-4 (inet::Packet, id=2363)
INFO: Sending app packet UDPData-4 over ipv4.
INFO: (MessageDispatcher)WirelessC.hostA.in: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipout, outGate
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #79 t=0.069032867984 WirelessC.hostA.ipv4.ip (Ipv4, id=83) on UDPData-4 (inet::Packet, id=2363)
INFO: Received (inet::Packet)UDPData-4 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
INFO: (Aodv)WirelessC.hostA.aodv: Finding route for source <unspec> with destination 10.0.0.2
INFO: (Aodv)WirelessC.hostA.aodv: Missing route for destination 10.0.0.2
DETAIL (Aodv)WirelessC.hostA.aodv: Queuing datagram, source <unspec>, destination 10.0.0.2
DETAIL (Aodv)WirelessC.hostA.aodv: Route discovery is in progress, originator 10.0.0.1 target 10.0.0.2
** Event #80 t=0.06901709225 WirelessC.hostA.app[0] (UdpBasicApp, id=58) on selfmsg sendTimer (inet::ClockEvent, id=2164)
INFO: (MessageDispatcher)WirelessC.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketout, out
** Event #81 t=0.06901709225 WirelessC.hostA.udp (Udp, id=55) on UDPData-5 (inet::Packet, id=2364)
INFO: Sending app packet UDPData-5 over ipv4.
INFO: (MessageDispatcher)WirelessC.hostA.in: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipout, outGate
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #82 t=0.06901709225 WirelessC.hostA.ipv4.ip (Ipv4, id=83) on UDPData-5 (inet::Packet, id=2364)
INFO: Received (inet::Packet)UDPData-5 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
INFO: (Aodv)WirelessC.hostA.aodv: Finding route for source <unspec> with destination 10.0.0.2
INFO: (Aodv)WirelessC.hostA.aodv: Missing route for destination 10.0.0.2
DETAIL (Aodv)WirelessC.hostA.aodv: Queuing datagram, source <unspec>, destination 10.0.0.2
DETAIL (Aodv)WirelessC.hostA.aodv: Route discovery is in progress, originator 10.0.0.1 target 10.0.0.2
** Event #83 t=0.071011024563 WirelessC.hostA.app[0] (UdpBasicApp, id=58) on selfmsg sendTimer (inet::ClockEvent, id=2164)
INFO: (MessageDispatcher)WirelessC.hostA.at: Dispatching packet to service, protocol = udp(75), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← app[0].socketout, out
** Event #84 t=0.071011024563 WirelessC.hostA.udp (Udp, id=55) on UDPData-6 (inet::Packet, id=2365)
INFO: Sending app packet UDPData-6 over ipv4.
INFO: (MessageDispatcher)WirelessC.hostA.in: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← udp.ipout, outGate
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.up: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← <parent>.trans
INFO: (MessageDispatcher)WirelessC.hostA.ipv4.mp: Dispatching packet to service, protocol = ipv4(50), servicePrimitive = 1, inGate = (omnetpp::cGate)in[0] ← up.out[2], out
** Event #85 t=0.071011024563 WirelessC.hostA.ipv4.ip (Ipv4, id=83) on UDPData-6 (inet::Packet, id=2365)
INFO: Received (inet::Packet)UDPData-6 (1008 B) (inet::SequenceChunk) length = 1008 B from upper layer.
INFO: (Aodv)WirelessC.hostA.aodv: Finding route for source <unspec> with destination 10.0.0.2
INFO: (Aodv)WirelessC.hostA.aodv: Missing route for destination 10.0.0.2
DETAIL (Aodv)WirelessC.hostA.aodv: Queuing datagram, source <unspec>, destination 10.0.0.2

```

Step 14 Introducing Antenna Gain

Antenna gain is introduced by replacing the default isotropic (omnidirectional) antenna with a directional antenna model (e.g., a dipole or a custom gain pattern). Antenna gain focuses transmitted power in specific directions, effectively increasing range in those directions while reducing it in others. This step demonstrates how physical-layer hardware choices directly impact network-level performance metrics like throughput and coverage area.





4. Simulation Results Summary

4.1 Key Metrics by Step

The table below tracks the key observable metric across selected steps. Fill in values from your scalar result files (.sca) after running each step.

Step	Feature Introduced	Packets Delivered	Key Observation
1	Basic wireless	1	Successful direct delivery using an ideal Unit Disk radio model.
3	Multi-hop topology	0	Delivery fails because hostA and hostB are out of range and no routing protocol is active.
4	Static routing	1	Multi-hop delivery succeeds as ipv4 routes packets through R1 and R2.
5	SINR interference	2	Realistic path loss and interference are introduced; packets may drop if signals overlap.
6	CSMA MAC	2	Nodes now "listen" before speaking, reducing collisions in the shared medium.
7	CSMA + ACKs	2	Reliability increases; lost packets are retransmitted at the MAC layer.
8	Energy model	3	Power consumption is now tracked for every radio state (TX/RX/Idle).
10	AODV routing	0*	(Discovery phase) Nodes broadcast RREQs to find paths dynamically instead of using static tables.
14	Antenna gain	0*	Focused signal strength increases range in specific directions but may limit others.

5. AI-Assisted Analysis

The simulation outputs from each major step were analysed using Claude AI. The analysis interprets observed behaviours, explains underlying networking principles, and compares performance across steps.⁴

5.2 AI Analysis of Results

The simulation results demonstrate a clear progression from an idealized communication model to a high-fidelity realistic environment. In the early steps, connectivity was binary: nodes either worked perfectly or not at all. The introduction of CSMA (Step 6) and ACKs (Step 7) highlights the transition to a coordinated medium where nodes manage contention, significantly reducing packet loss at the cost of slight protocol overhead.

As we moved to Step 10 (AODV), the simulation shifted from static "wired-like" logic to true ad-hoc networking, where the network autonomously repairs itself as nodes move. The final steps (12-14) moved the simulation into the physical domain, where the dimensional analog model and antenna gain showed that network performance is not just about logic, but about the physical environment and hardware capabilities. Higher-fidelity models in Step 12 showed more realistic packet corruption, demonstrating why real-world Wi-Fi often sees fluctuating throughput.

6. Conclusion

This lab demonstrated the incremental design of a wireless network simulation using OMNeT++ and the INET Framework. By progressing through 14 steps, we observed the following key principles:

- A functional wireless simulation requires coordinated layers: physical (radio model), MAC (CSMA), network (routing), and application.
- Ideal radio models (UnitDiskRadio) are sufficient for basic connectivity testing but mask real interference effects.
- CSMA significantly reduces collisions in shared wireless media; adding ACKs further improves reliability at the cost of overhead.
- Static routing breaks immediately when nodes are mobile; AODV restores connectivity through on-demand route discovery.
- Physical obstacles, realistic path loss models, and antenna gain all significantly affect coverage and throughput in ways that simple models cannot capture.

"I specifically observed that as we added more realistic constraints such as interference in Step 5 and physical obstacles in Step 11 the simple connectivity we saw in Step 1 became much harder to maintain. This underscored the importance of robust protocols like AODV and CSMA in ensuring reliability in unpredictable wireless environments."

7. References

- INET Framework Wireless Tutorial: <https://inet.omnetpp.org/docs/tutorials/wireless/doc/index.html>
- OMNeT++ Documentation: <https://omnetpp.org/documentation/>
- INET 4.6.0 User's Guide: <https://inet.omnetpp.org/docs/users-guide/index.html>
- Haliç University — Computer Networks Course, Lab 3 Assignment, 2026